25DCE097

DEPSTAR

Devang Patel Institute of Advance Technology and Research

(A Constitute Institute of CHARUSAT)

Certificate

This is to certify that

~~Mr.~~/ Mrs. Rajyani Drashti N.

of Depstar-CE *Class,*

ID. No. 25DCE097 *has satisfactorily completed his/ her term work in* ITUE203 - web development for and framework

the ending in June/July *2026 /2027*

Date : 21/10/26

22/10/2026

Sign. of Faculty

Head of Department

B2

DEPSTAR

Devang Patel Institute of Advance Technology and Research

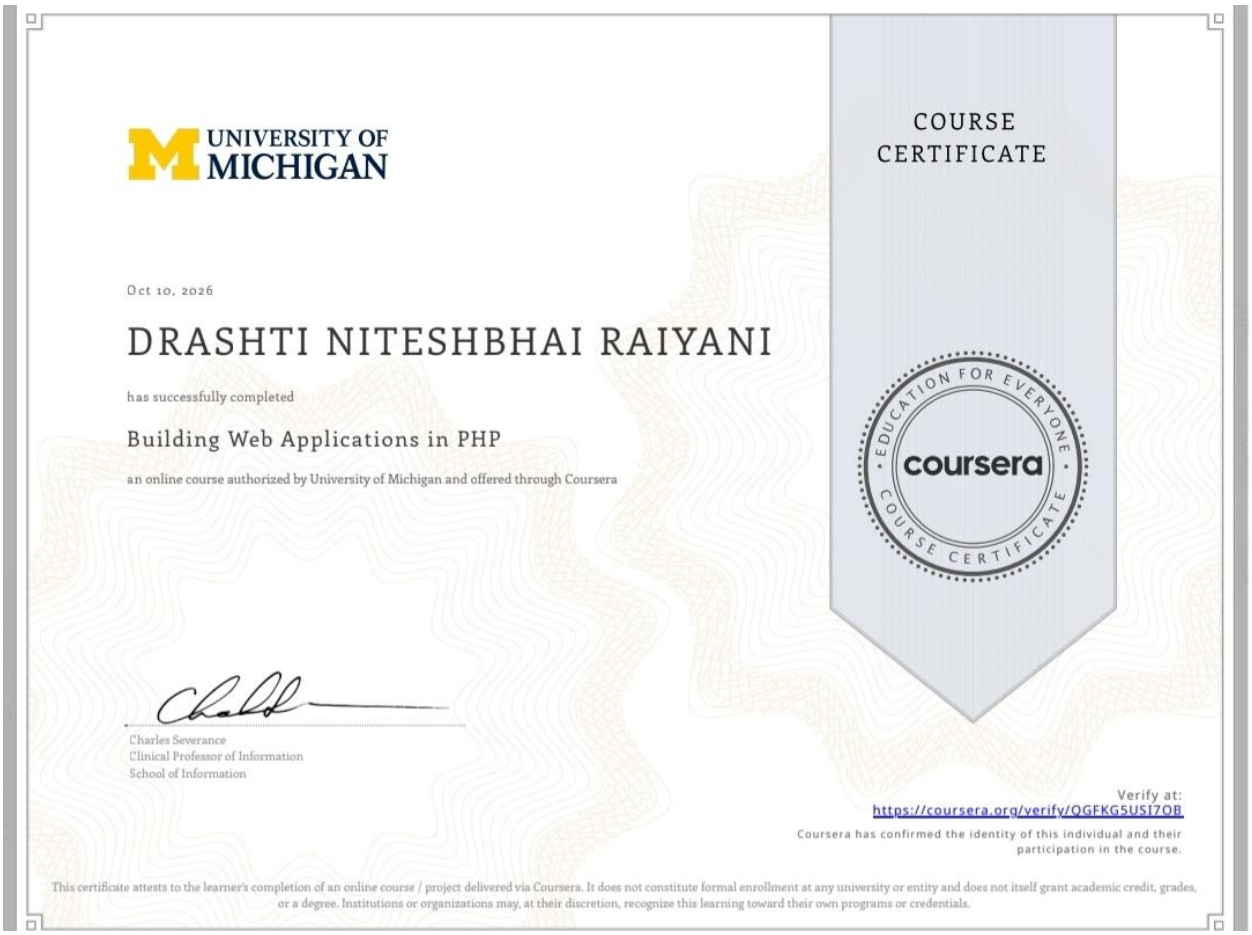
(A Constitute Institute of CHARUSAT)

Name : Raiyani Darshti Niteshbhai ID. No. 25DCE097

INDEX

No.	DATE	SUBJECT	PAGE	MARKS	SIGNATURE
1.	08/07/'26	Practical - 1	03		
2.	15/07/'26	Practical - 2	08		
3.	16/07/'26	Practical - 3	25		
4.	22/07/'26	Practical - 4	45		
5.	29/07/'26	Practical - 5	48		
6.	05/08/'26	Practical - 6	53		
7.	06/08/'26	Practical - 7	56		
8.	12/08/'26	Practical - 8	60		
9.	19/08/'26	Practical - 9	64		
10.	20/08/'26	Practical - 10	79		
11.	26/08/'26	Practical - 11	82		
12.	27/08/'26	Practical - 12	89		
13.	02/09/'26	Practical - 13	98		
14.	16/09/'26	Practical - 14	106		
15.	29/09/'26	Practical - 15	108		

ITUE:203-WDF **25DCE097**



UNIVERSITY OF MICHIGAN

COURSE CERTIFICATE

Oct 10, 2026

DRASHTI NITESHBHAI RAIYANI

has successfully completed

Building Web Applications in PHP

an online course authorized by University of Michigan and offered through Coursera

EDUCATION FOR EVERYONE · coursera · COURSE CERTIFICATE

Charles Severance
Clinical Professor of Information
School of Information

Verify at:
https://coursera.org/verify/QGFKG5USI7OB

Coursera has confirmed the identity of this individual and their participation in the course.

This certificate attests to the learner's completion of an online course / project delivered via Coursera. It does not constitute formal enrollment at any university or entity and does not itself grant academic credit, grades, or a degree. Institutions or organizations may, at their discretion, recognize this learning toward their own programs or credentials.

THE CSS BOX MODEL

More than just a box — it's the foundation of your layout!

Style Structure Better Web

1. WHAT IS THE BOX MODEL?

The CSS Box Model is a fundamental concept that describes how elements are structured and spaced on a web page. Every element is treated as a rectangular box, consisting of:

Content + Padding + Border + Margin

It helps control the size, spacing and layout of elements, making your web pages well-organized and visually appealing.

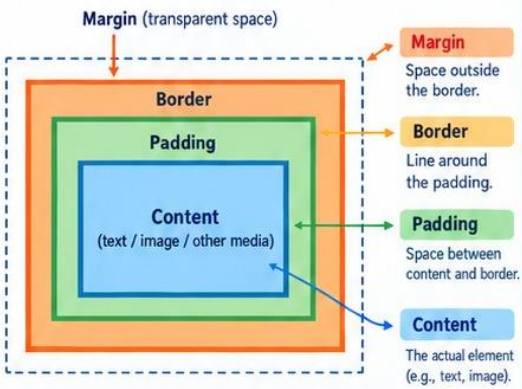
2. THE PROBLEM / REQUIREMENT

Why is it important?

Without understanding the box model, elements may overflow, appear misaligned, or create inconsistent spacing. This leads to poor layout, broken designs and a bad user experience.

The box model solves this by giving developers precise control over the dimensions and spacing of elements.

3. THE BOX MODEL ANATOMY



Total Element Width / Height

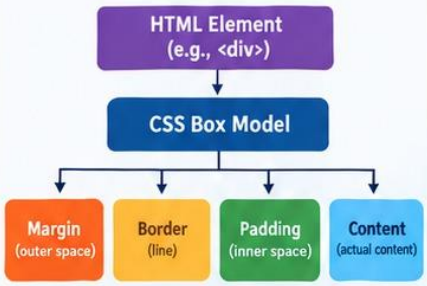
= Content + Padding + Border + Margin

```
width: content + padding + border + margin;
```

4. KEY FEATURES

- **Control Over Size** — Use width, height, padding, border and margin.
- **Flexible Layouts** — Works with block, inline and inline-block elements.
- **Responsive Design** — Helps create consistent layouts across devices.
- **Visual Spacing** — Improves readability and design, balance.
- **Easy to Debug** — Chrome DevTools helps inspect and modify box model properties in real time.

5. ARCHITECTURE & COMPONENTS



The browser calculates the total size based on these four components.

6. HOW IT WORKS (REQUEST – RESPONSE FLOW)

1. **User writes HTML + CSS** — Defines the structure and styles.
2. **Browser parses the code** — Builds the DOM and applies CSS rules.
3. **Box model is applied** — Calculates size, spacing and layout of each element.
4. **Page is rendered** — User sees the styled web page.

7. IMPLEMENTATION EXAMPLE

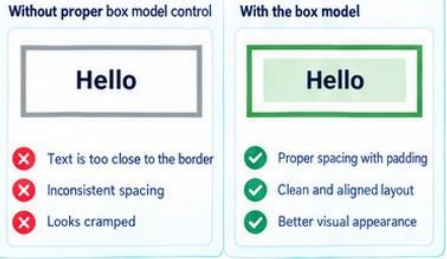
HTML

```html
<div class="box">
  Hello, CSS Box Model!
</div>
```

CSS

```css
.box {
  width: 200px;
  padding: 20px;
  border: 5px solid #4CAF50;
  margin: 30px;
  background-color: #e8f5e9;
  text-align: center;
  font-family: Arial, sans-serif;
}
```

8. VISUAL EXAMPLE – BEFORE & AFTER



9. REAL-WORLD EXAMPLES



10. DESIGN CONSIDERATIONS & BEST PRACTICES

- Use consistent spacing (e.g., rem, em, px).
- Consider the box-sizing property (border-box is often better).
- Avoid hardcoding sizes; use responsive units (%, vw, vh, etc.).
- Use DevTools to inspect and debug layout issues.
- Keep your CSS clean, modular and reusable.

11. KEY TAKEAWAYS

- The box model = content + padding + border + margin.
- It controls size, spacing and layout.
- It's essential for responsive and consistent design.
- Mastering it leads to better, cleaner and more professional web applications.

Understand the Box Model → Build Better Layouts → Create Amazing Web Apps

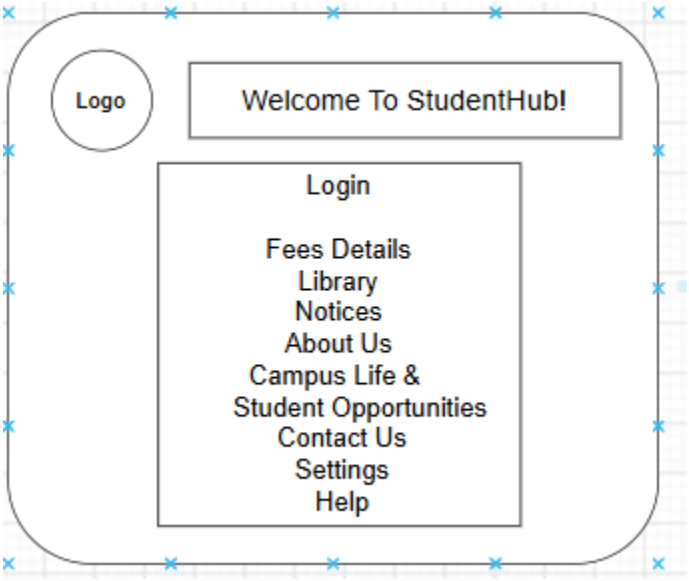
Shah Devanshi - 25DCE107 | Raiyani Drashti - 25DCE097

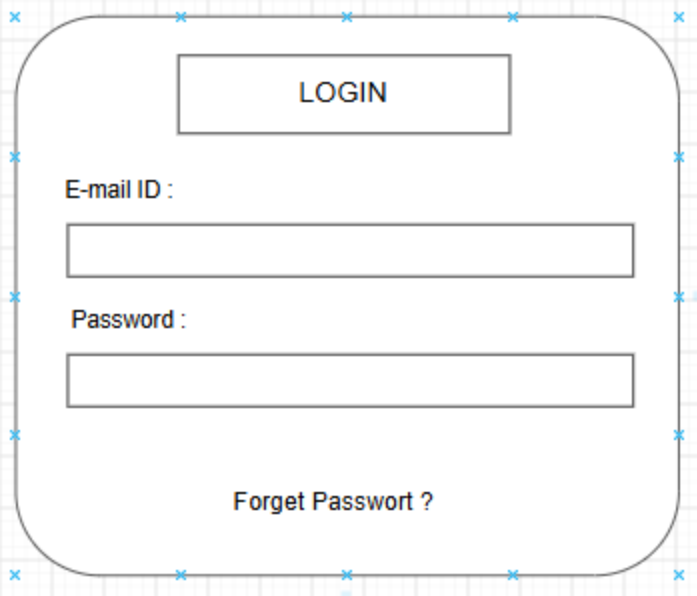
Practical 1: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

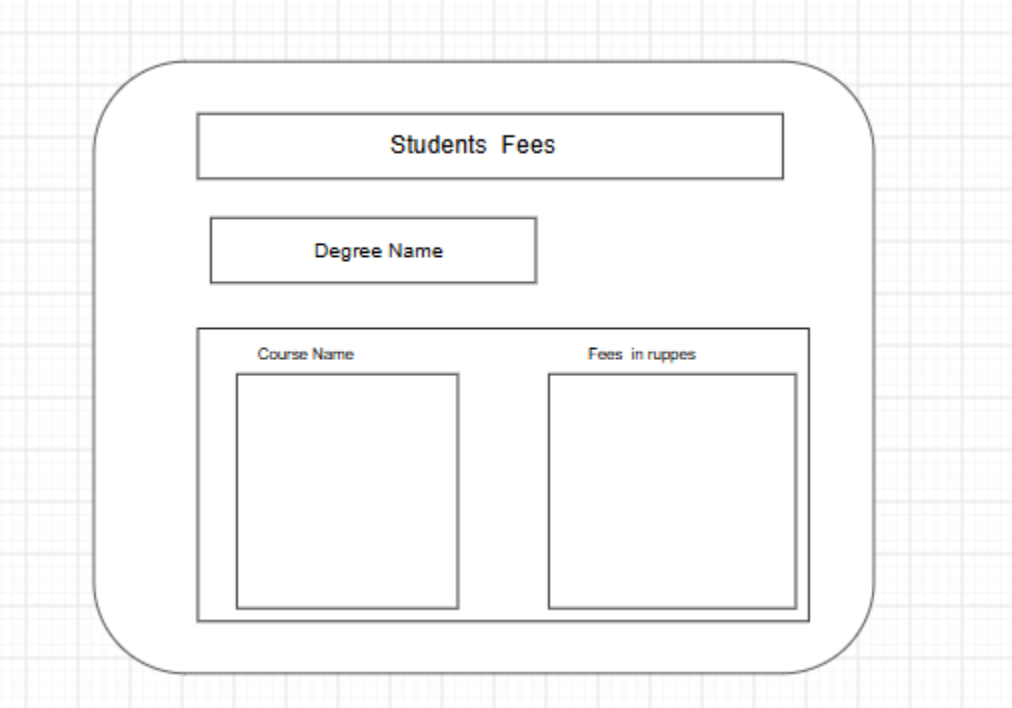
Problem Definition Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and

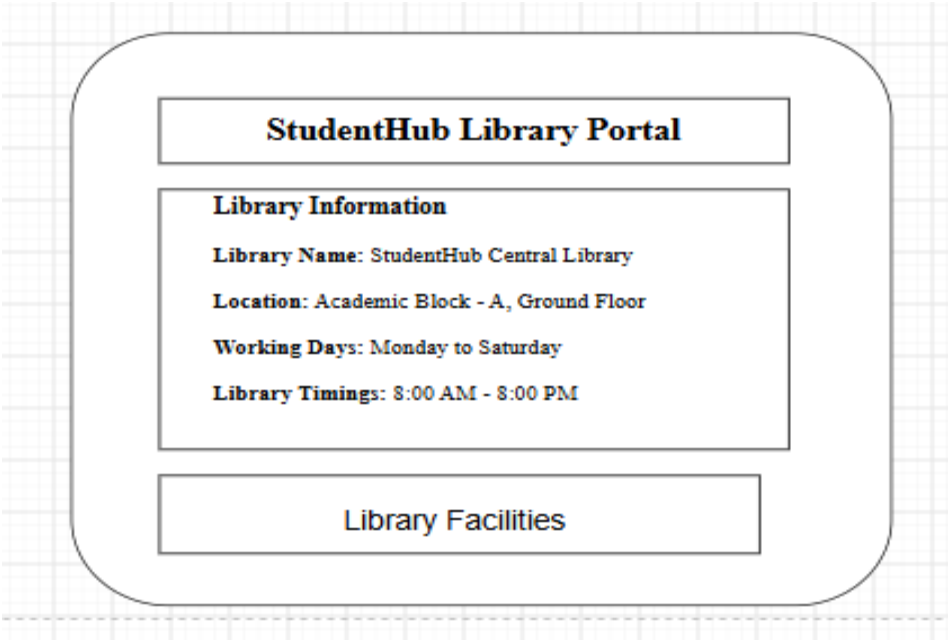
GitHub repository.The Code for This all the output Screenshots is uploaded on Github :

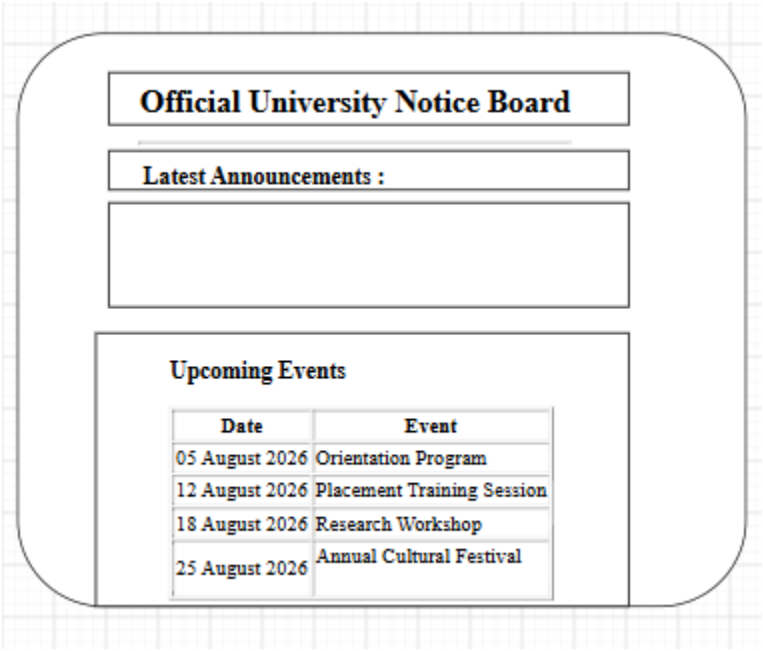
https://github.com/raiyanidrashti21-bit/25dce097-Web



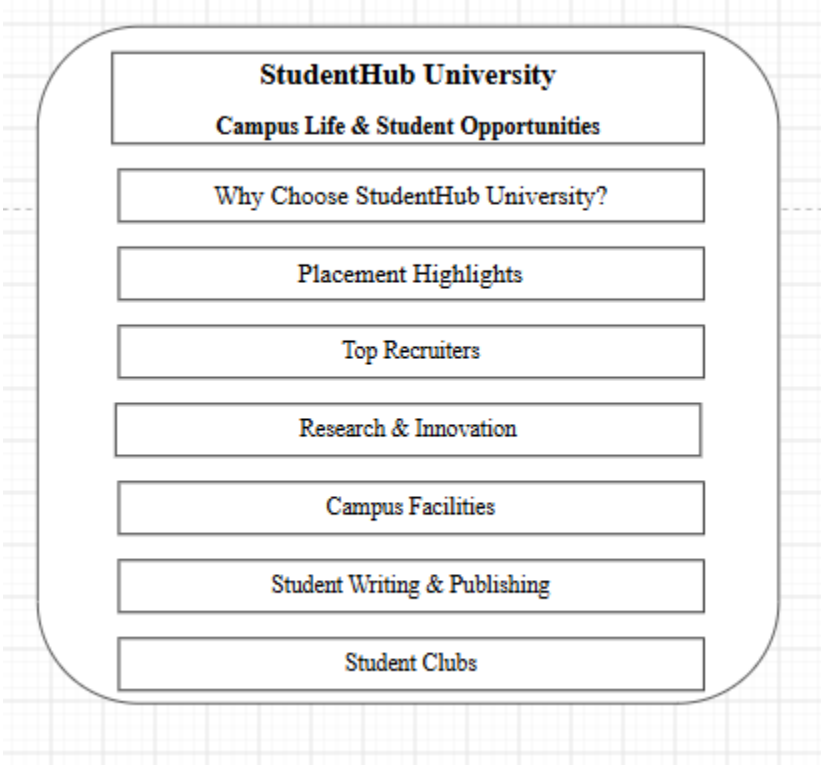

LOGIN
E-mail ID :
Password :
Forget Passwort ?

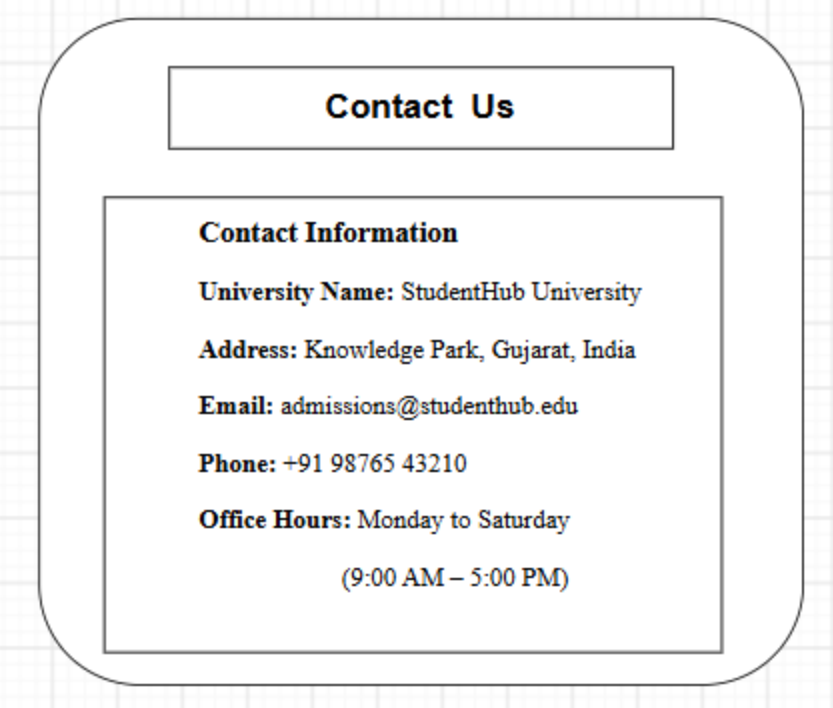

Students Fees
Degree Name
Course Name
Fees in ruppes

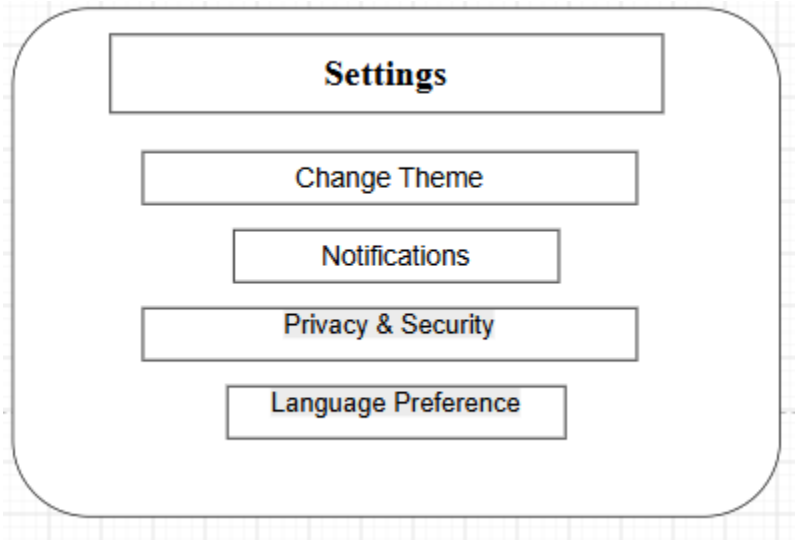

StudentHub Library Portal
Library Information
Library Name: StudentHub Central Library
Location: Academic Block - A, Ground Floor
Working Days: Monday to Saturday
Library Timings: 8:00 AM - 8:00 PM
Library Facilities


Official University Notice Board
Latest Announcements :
Upcoming Events
Date
Event
05 August 2026
Orientation Program
12 August 2026
Placement Training Session
18 August 2026
Research Workshop
25 August 2026
Annual Cultural Festival


About University
Our Vision
Our Mission
Academic Programs


StudentHub University
Campus Life & Student Opportunities
Why Choose StudentHub University?
Placement Highlights
Top Recruiters
Research & Innovation
Campus Facilities
Student Writing & Publishing
Student Clubs





Practical 2

Semantic HTML5 Pages with Accessibility-Ready Structure

Problem Definition

Develop static HTML5 skeletons for at least 10 StudentHub pages such as Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback. Use semantic HTML5 tags and accessibility-friendly structures to create clean, consistent, and easy-to-navigate web pages.

In this practical, static HTML5 structures were created for the StudentHub portal using semantic HTML5 elements such as header, nav, main, section, article, aside, and footer. Ten pages including Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, FAQ, and Feedback were linked together using navigation links. Accessibility features such as proper form labels, heading hierarchy, alternative text for images, and meaningful navigation links were used to make the pages easier to understand and navigate. A consistent structure was maintained across all pages.

Questions:

1. Are semantic tags such as header, nav, main, section, article, aside, and footer used properly?

Yes, semantic HTML5 tags were used properly to organize the webpage structure. The header contains the page heading and navigation, nav contains the navigation links, main contains the primary content, section and article are used to divide related content, aside is used for additional information, and footer contains footer information. This makes the page structure clear and meaningful.

2. Are form labels, headings, image alt text, and navigation links accessible?

Yes, form fields were provided with proper labels, headings were arranged in a logical hierarchy, and images were given suitable alt text. Navigation links were given meaningful names so users can understand their purpose. These practices improve accessibility and make the website easier to use with screen readers and other accessibility tools.

3. Is the page structure consistent across all pages?

Yes, a consistent structure was maintained across all StudentHub pages. Common elements such as the header, navigation menu, main content area, and footer were used throughout the pages. This provides a similar navigation experience and makes the website easier for users to understand.

Dashboard :

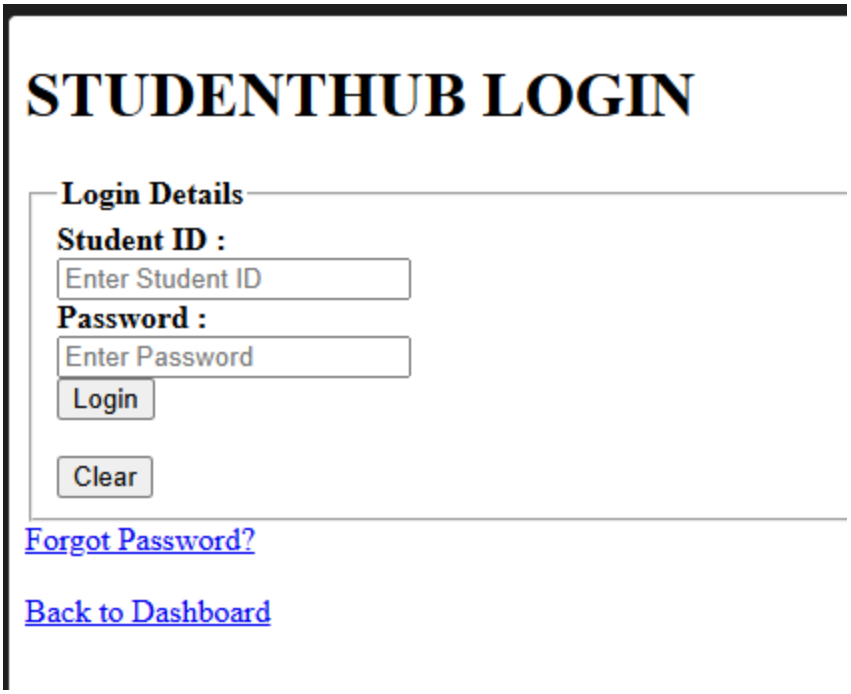
StudentHub

Student Portal Dashboard

- Login
- Fees Details
- Library
- Notices
- About Us
- Campus Life & Student Opportunities
- Contact Us
- Settings
- Help

@ 2026 StudentHub

Login :


STUDENTHUB LOGIN
Login Details
Student ID :
Enter Student ID
Password :
Enter Password
Login
Clear
Forgot Password?
Back to Dashboard

Fees :

StudentHub University

Academic Fee Structure (Sample)

Academic Year : 2026-2027

Engineering Programs

Course	Annual Tuition Fee	NRI Fee
B.Tech Computer Engineering	₹1,45,000	₹2,90,000
B.Tech Information Technology	₹1,45,000	₹2,90,000
B.Tech Artificial Intelligence & Data Science	₹1,45,000	₹2,90,000
B.Tech Mechanical Engineering	₹1,45,000	₹2,90,000
B.Tech Civil Engineering	₹1,45,000	₹2,90,000
B.Tech Electrical Engineering	₹1,45,000	₹2,90,000
B.Tech Electronics & Communication	₹1,45,000	₹2,90,000

Health Science Programs

Course	Annual Tuition Fee	NRI Fee
B.Pharmacy	₹1,20,000	₹2,40,000
Doctor of Pharmacy (Pharm.D)	₹1,75,000	₹3,50,000
BPT (Physiotherapy)	₹1,10,000	₹2,20,000

Activate Windows
Go to Settings to activate Wi

B.Optometry	₹95,000	₹1,90,000
B.Sc Nursing	₹1,15,000	₹2,30,000

Commerce & Management

Course	Annual Tuition Fee	NRI Fee
BBA	₹75,000	₹1,50,000
B.Com	₹55,000	₹1,10,000
MBA	₹1,30,000	₹2,60,000

Computer Applications

Course	Annual Tuition Fee	NRI Fee
BCA	₹85,000	₹1,70,000
MCA	₹1,10,000	₹2,20,000

Science & Arts

Course	Annual Tuition Fee	NRI Fee
B.Sc	₹60,000	₹1,20,000
M.Sc	₹70,000	₹1,40,000
BA	₹45,000	₹90,000
MA	₹55,000	₹1,10,000

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Payment Information

- Fees can be paid online through the StudentHub Portal.
- Students can pay semester-wise or annually.
- NRI fees are approximately double the regular tuition fee.
- Hostel, transport, examination and library fees are charged separately.
- Scholarships are available for eligible students according to university rules.

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Back to Dashboard

Library :

StudentHub Library Portal

Library Information

Library Name: StudentHub Central Library

Location: Academic Block - A, Ground Floor

Working Days: Monday to Saturday

Library Timings: 8:00 AM - 8:00 PM

Digital Library Services

- E-Books Collection
- E-Journals
- Research Papers
- Previous Year Question Papers
- Online Thesis Repository
- IEEE Digital Library Access
- NPTEL Video Lectures
- Remote Digital Library Access

Computer & Internet Section

- 50 Desktop Computers
- High-Speed Internet Access
- Printing and Scanning Facility
- Online Examination Practice
- Programming Software Installed
- Wi-Fi Available

Reading Hall

- Individual Reading Tables
- Silent Study Zone
- Air Conditioned Reading Room
- Newspaper & Magazine Section
- Open from 8:00 AM to 8:00 PM

Group Discussion & Project Area

- Project Discussion Rooms
- Group Study Tables
- Whiteboard Facility
- Project Presentation Area
- Meeting Rooms (Advance Booking Required)

Library Rules

1. Carry Student ID Card.
2. Maintain Silence.
3. No Food or Drinks Allowed.
4. Return Books Before Due Date.
5. Handle Books Carefully.
6. Mobile Phones Must Be Silent.

Library Contact

Librarian: Dr. Meera Shah

Email: library@studenthub.edu

Phone: +91 9876543210

Back to Dashboard

Notices :

StudentHub University

Official University Notice Board

Latest Announcements

- Academic Calendar for the current semester has been published.
- Online Examination Registration Portal is now open.
- Scholarship applications are available for eligible students.
- Library membership renewal has started.
- University Convocation registration is now open.
- Campus Placement Drive schedule has been updated.
- National Workshop on Artificial Intelligence will be held next month.
- Research paper submission guidelines have been updated.
- Annual Sports Festival registrations are now open.
- Cultural Festival event schedule has been published.

Upcoming Events

Date	Event
05 August 2026	Orientation Program
12 August 2026	Placement Training Session
18 August 2026	Research Workshop
25 August 2026	Annual Cultural Festival
05 September 2026	Industry Interaction Session
15 September 2026	Innovation & Startup Seminar

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Important Links

- Academic Calendar
- Examination Schedule
- Scholarship Information
- Placement Cell Updates
- Library Announcements
- University Circulars
- Research Notices
- Student Activity Updates
- Holiday List
- Download Official Forms

Back to Dashboard

About Us:

StudentHub University

About Us

Our Vision

StudentHub University aims to become a globally recognized institution by providing quality education, encouraging innovation, promoting research, and preparing students to become responsible professionals and future leaders.

Our Mission

- Provide quality education through experienced faculty.
- Promote research, innovation and entrepreneurship.
- Develop leadership and communication skills.
- Strengthen collaboration with industries and organizations.
- Encourage ethical values and social responsibility.
- Create lifelong learning opportunities for students.

Academic Programs

StudentHub University offers undergraduate, postgraduate and doctoral programs in various disciplines to prepare students for successful careers.

- Engineering & Technology
- Computer Applications
- Management Studies
- Commerce
- Science
- Health Sciences
- Arts & Humanities
- Pharmacy

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Explore More

To know more about placements, campus facilities, sports, research, student clubs, cultural activities, international collaborations and student support services, visit the Campus Life page.

Campus Life & Student Opportunities

Back to Dashboard

Activate Winc
Go to Settings to a

Campus Life :

StudentHub University

Official University Notice Board

Stay updated with the latest university announcements, examination notices, placement drives, scholarships, events, and important academic information.

Latest Announcements

- Academic Calendar for Semester 2026-27 has been published.
- Online Examination Registration Portal is now open.
- Library Membership Renewal has started.
- Scholarship Applications are available for eligible students.
- University Convocation Registration is now open.
- Campus Placement Drive schedule has been updated.
- National Workshop on Artificial Intelligence will be conducted next month.
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12 August 2026	Placement Training Session
18 August 2026	Research & Innovation Workshop
25 August 2026	Annual Cultural Festival
05 September 2026	Industry Interaction Session
15 September 2026	Startup & Entrepreneurship Seminar

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Examination Notices

Notice	Status
Semester Examination Time Table	Available
Online Exam Registration	Open
Hall Ticket Download	Available Soon
Rechecking & Reassessment Forms	Closed
Result Declaration	To Be Announced

Admission Updates

- Admissions for Academic Year 2026-27 are Open.
- Merit List will be published on 10 August 2026.
- Document Verification Schedule Released.
- Hostel Admission Forms Available.
- Fee Payment Last Date: 20 August 2026.

ITUE:203-WDF **25DCE097**

Placement Cell Updates

Company	Drive Date
TCS	22 August 2026
Infosys	28 August 2026
Wipro	05 September 2026
LTIMindtree	10 September 2026
HCL Technologies	18 September 2026

Scholarships & Financial Aid

- Government Scholarship Applications Open.
- Merit Scholarships for University Rank Holders.
- Sports Scholarships Available.
- Financial Assistance for Economically Weaker Students.
- Last Date for Scholarship Submission: 30 August 2026.

Library Updates

- Digital Library Portal Updated.
- New Engineering, Medical and Commerce Books Added.
- Extended Reading Hall Timings During Examination.
- Research Journal Access Available.
- Library Open from 8:00 AM to 10:00 PM.

Quick Information

Service	Status
Academic Calendar	Available
Examination Schedule	Available
Library Portal	Open
Fee Payment Portal	Open
Student Help Desk	Monday - Saturday

Important Links

- Academic Calendar
- Examination Schedule
- Scholarship Information
- Placement Cell Updates
- Library Announcements
- Research Notices
- Holiday List
- Download Official Forms

Back to Dashboard

Contact Us :

Contact Us

Email : studenthub@gmail.com

Phone : +91 9876543210

Address : CHARUSAT University, Changa, Gujarat

Back to Dashboard

Settings :

StudentHub University

Settings

My Profile

Change Password

Notification Settings

Privacy & Security

Language Preferences

← Back to Dashboard

Help :

Help Center

For any technical issue or academic support, please contact the StudentHub Help Desk.

Email : help@studenthub.com

Phone : +91 9999999999

Back to Dashboard

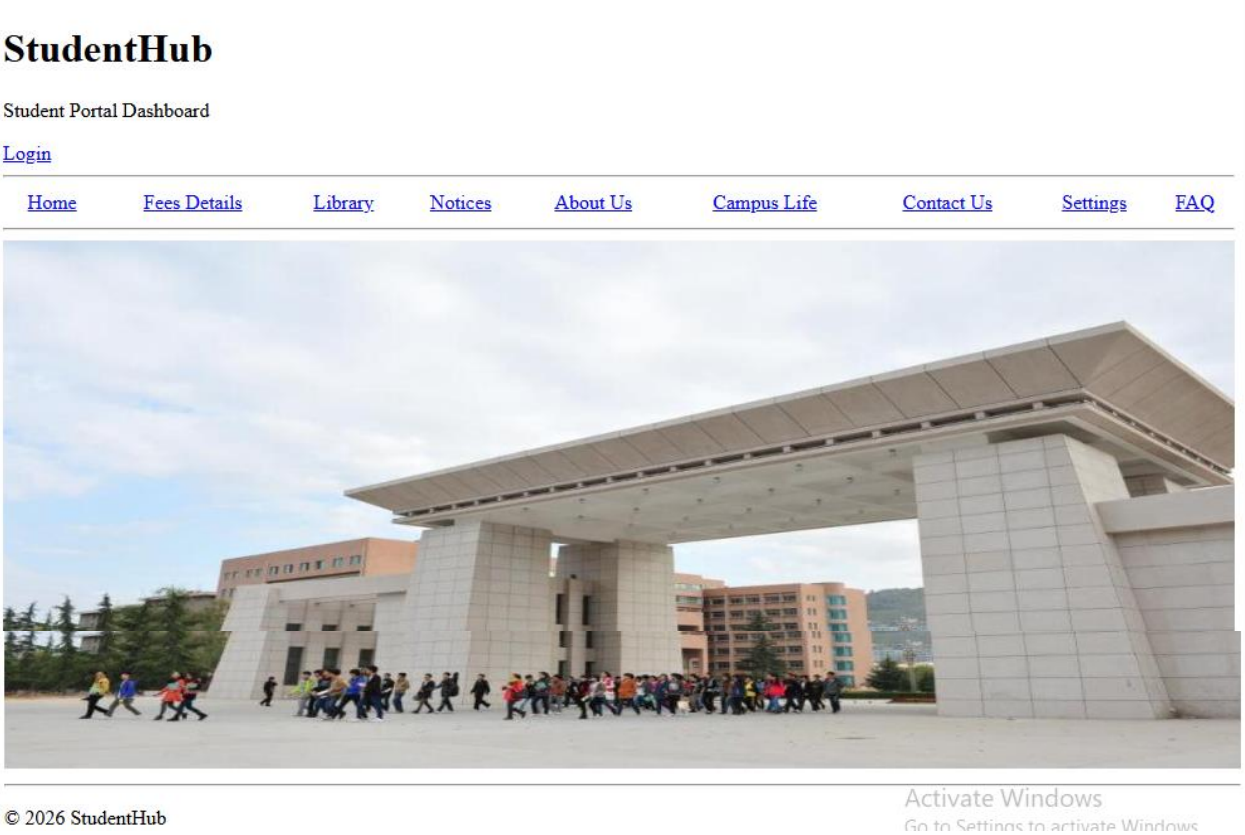
Learning Outcomes:

Students were able to create clean and structured StudentHub web pages using semantic HTML5 elements. They learned to use header, nav, main, section, article, aside, and footer correctly, create accessible forms with proper labels, add alt text to images, and link multiple pages through navigation. They also understood the importance of consistent page structure and basic accessibility practices.

Conclusion:

The practical successfully demonstrated how semantic HTML5 and accessibility features can be used to create well-structured and user-friendly StudentHub pages. The use of proper semantic tags, accessible forms, meaningful navigation links, and consistent page layouts makes the website easier to understand, navigate, and maintain.

Changes That I made :



The Students Dashboard:

Welcome, Student!

Welcome to your StudentHub personal dashboard.

Student Profile

View your personal and academic information.

Attendance Report

Check your subject-wise attendance.

Time Table

View your weekly class schedule.

Examination / Results

Check examination information and results.

← Back to Main Dashboard

←

Class Time Table

Academic Year: 2026-2027

Day	9:00 - 10:00	10:00 - 11:00	11:00 - 12:00	12:00 - 1:00	2:00 - 3:00	3:00 - 4:00
Monday	OOP	Computer Networks	DBMS	Break	Web Development	Software Engineering
Tuesday	DBMS	OOP	Web Development	Break	Computer Networks Lab	
Wednesday	Software Engineering	DBMS	Computer Networks	Break	OOP Lab	OOP Lab
Thursday	Web Development	Computer Networks	OOP	Break	DBMS Lab	DBMS Lab
Friday	Computer Networks	Software Engineering	DBMS	Break	Web Development	Library

←

Examination Results

Academic Year: 2026-2027

Subject	Internal Marks	External Marks	Total	Grade
Object Oriented Programming	25	62	87	A
Computer Networks	24	60	84	A
Database Management	26	58	84	A
Web Development	23	61	84	A
Software Engineering	24	59	83	A

Semester Summary

Semester: 6

SGPA: 8.40

Result: PASS

←

Attendance Report

Academic Year: 2026-2027

Subject	Total Classes	Classes Attended	Attendance
Object Oriented Programming	40	36	90%
Computer Networks	38	33	87%
Database Management	42	37	88%
Web Development	40	34	85%
Software Engineering	36	32	89%

Overall Attendance

Your overall attendance is **88%**.

⇐

Student Profile

Student Personal and Academic Information

Personal Information

Student ID	25DCE097
Name	Student Name
Email	student@studenthub.com
Contact Number	9876543210
Date of Birth	01 December 2006

Academic Information

Course	B.Tech Computer Engineering
Semester	3rd Semester
Academic Year	2026-2027
Division	2
Batch	B

Practical 3

Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition

Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.Summary:

In this practical, responsive user interfaces were designed for the Home, About, Registration, Dashboard, and Events pages. CSS Grid and Flexbox were used to arrange the content and create layouts that adjust according to different screen

sizes. The pages were designed using a mobile-first approach so that they work properly on mobile, tablet, and desktop devices. Proper typography, spacing, colors, and contrast were also maintained to make the interface simple, readable, and user-friendly.

Questions:

1. How does the layout adapt for mobile, tablet, and desktop screens?

The layout adapts using CSS Grid, Flexbox, and media queries. On mobile screens, the content is arranged mostly in a single column. On tablets, the layout can use two columns, while on desktop screens, multiple columns and wider sections are displayed. This makes the website responsive and prevents content from overflowing.

2. Which layout technique is used and why?

CSS Flexbox and CSS Grid were used. Flexbox was used for arranging elements in a row or column and for proper alignment. CSS Grid was used for creating structured layouts with rows and columns, especially for dashboard and event sections. Both techniques make the layout easier to manage and responsive.

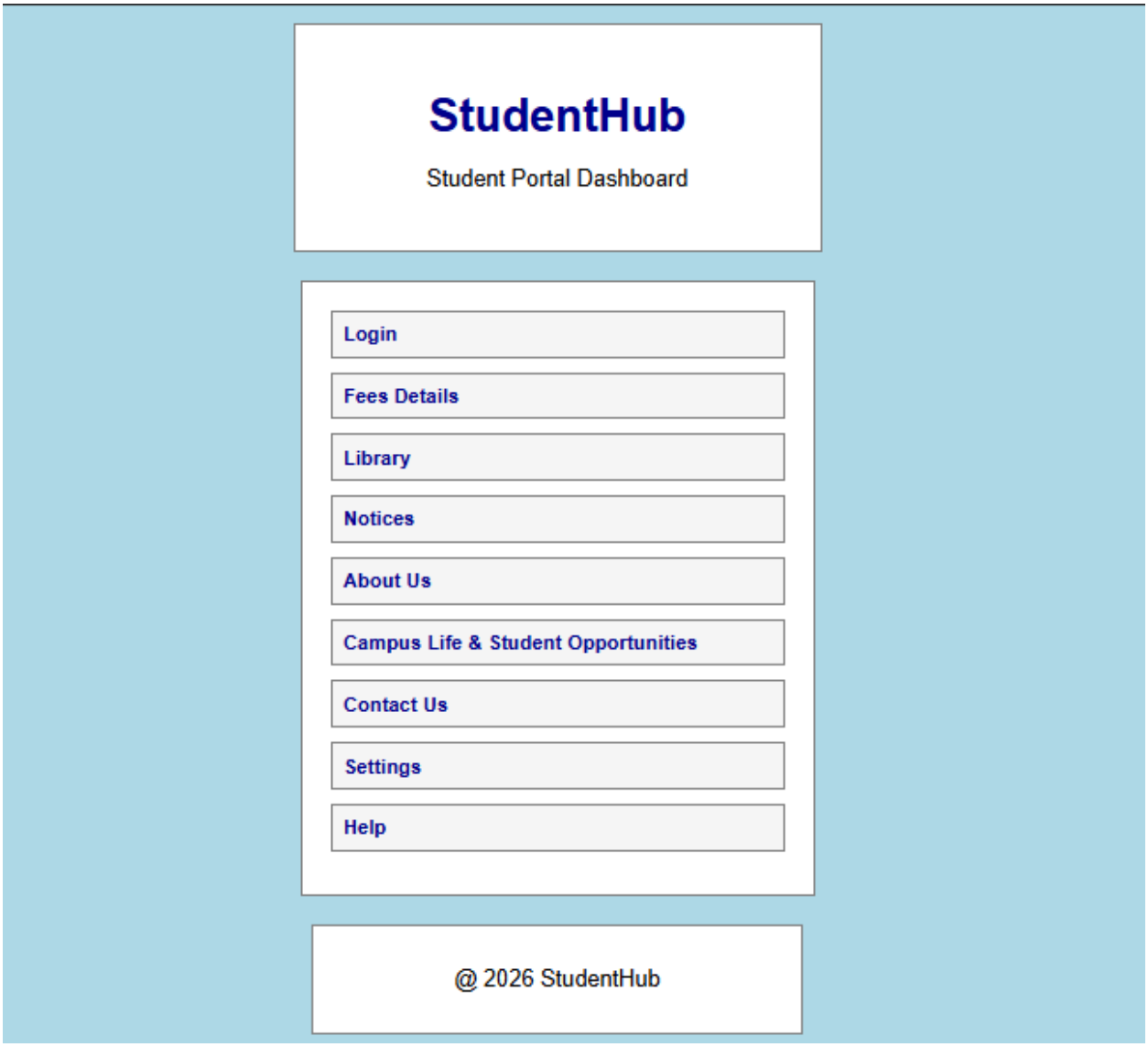
3. Are typography, spacing, colors, and contrast readable?

Yes, proper typography, spacing, colors, and contrast were maintained to improve readability. Clear font sizes were used for headings and normal text, sufficient spacing was provided between elements, and suitable colors with good contrast were selected so that the content is easy to read on different screen sizes.

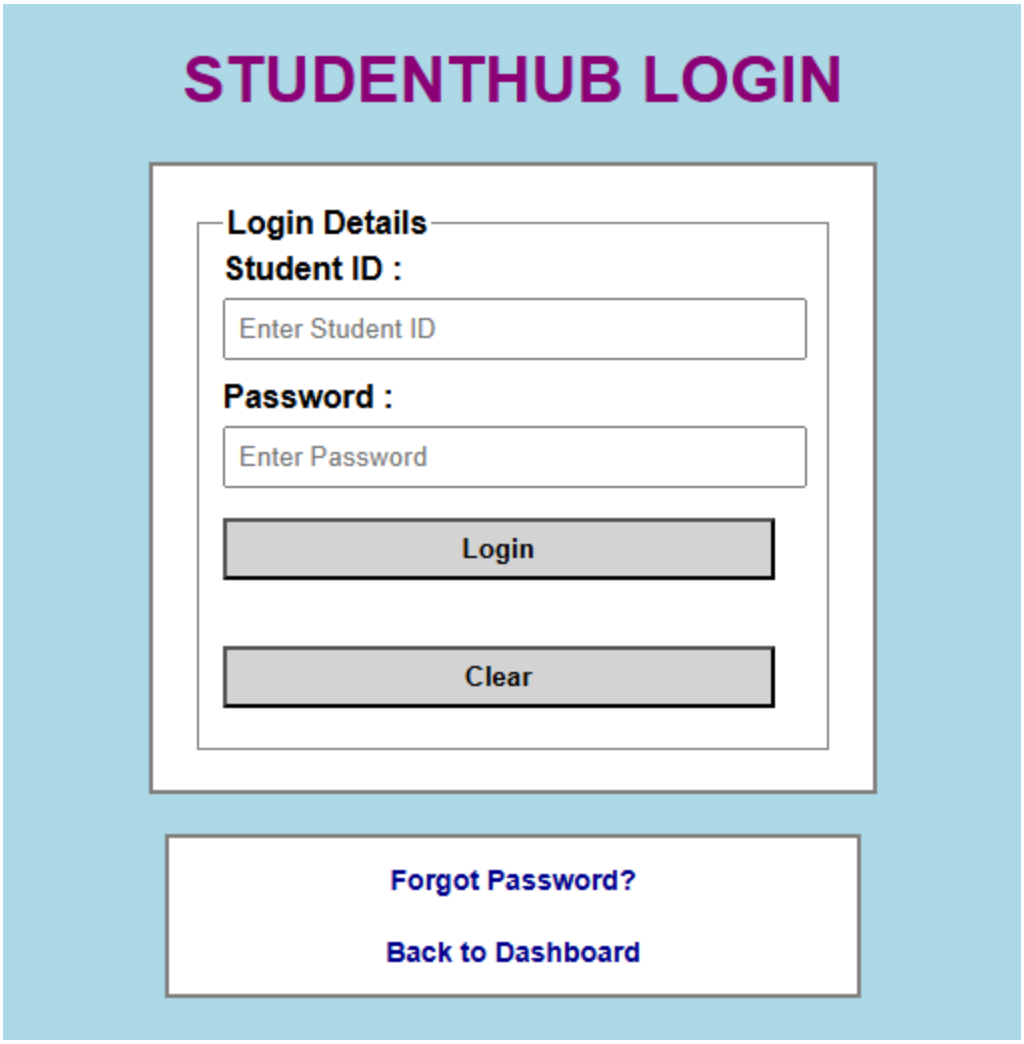
4. Is the framework used consistently without unnecessary inline CSS?

Yes, the framework was used consistently wherever required. Common classes and external CSS were preferred instead of unnecessary inline CSS. This keeps the HTML code clean, makes the design consistent across all pages, and makes future changes easier.

Dashboard :


StudentHub
Student Portal Dashboard
Login
Fees Details
Library
Notices
About Us
Campus Life & Student Opportunities
Contact Us
Settings
Help
@ 2026 StudentHub

Login :


STUDENTHUB LOGIN
Login Details
Student ID :
Enter Student ID
Password :
Enter Password
Login
Clear
Forgot Password?
Back to Dashboard

Fees :

ITUE:203-WDF **25DCE097**

StudentHub University

Academic Fee Structure (Sample)

Academic Year : 2026-2027

Engineering Programs

Course	Annual Tuition Fee	NRI Fee
B.Tech Computer Engineering	₹1,45,000	₹2,90,000
B.Tech Information Technology	₹1,45,000	₹2,90,000
B.Tech Artificial Intelligence & Data Science	₹1,45,000	₹2,90,000
B.Tech Mechanical Engineering	₹1,45,000	₹2,90,000
B.Tech Civil Engineering	₹1,45,000	₹2,90,000
B.Tech Electrical Engineering	₹1,45,000	₹2,90,000
B.Tech Electronics & Communication	₹1,45,000	₹2,90,000

Health Science Programs

Course	Annual Tuition Fee	NRI Fee
B.Pharmacy	₹1,20,000	₹2,40,000
Doctor of Pharmacy (Pharm.D)	₹1,75,000	₹3,50,000
BPT (Physiotherapy)	₹1,10,000	₹2,20,000
B.Optometry	₹95,000	₹1,90,000
B.Sc Nursing	₹1,15,000	₹2,30,000

Commerce & Management

Course	Annual Tuition Fee	NRI Fee
BBA	₹75,000	₹1,50,000
B.Com	₹55,000	₹1,10,000
MBA	₹1,30,000	₹2,60,000

ITUE:203-WDF **25DCE097**

Computer Applications

Course	Annual Tuition Fee	NRI Fee
BCA	₹85,000	₹1,70,000
MCA	₹1,10,000	₹2,20,000

Science & Arts

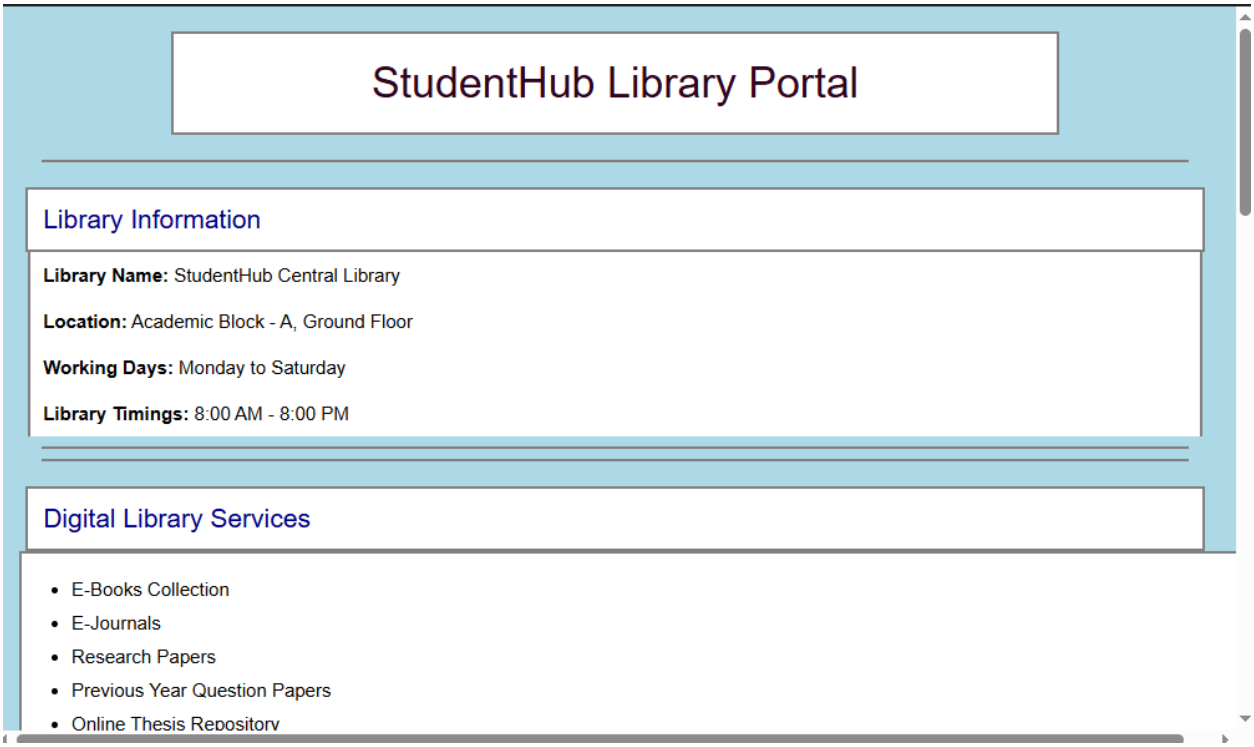
Course	Annual Tuition Fee	NRI Fee
B.Sc	₹60,000	₹1,20,000
M.Sc	₹70,000	₹1,40,000
BA	₹45,000	₹90,000
MA	₹55,000	₹1,10,000

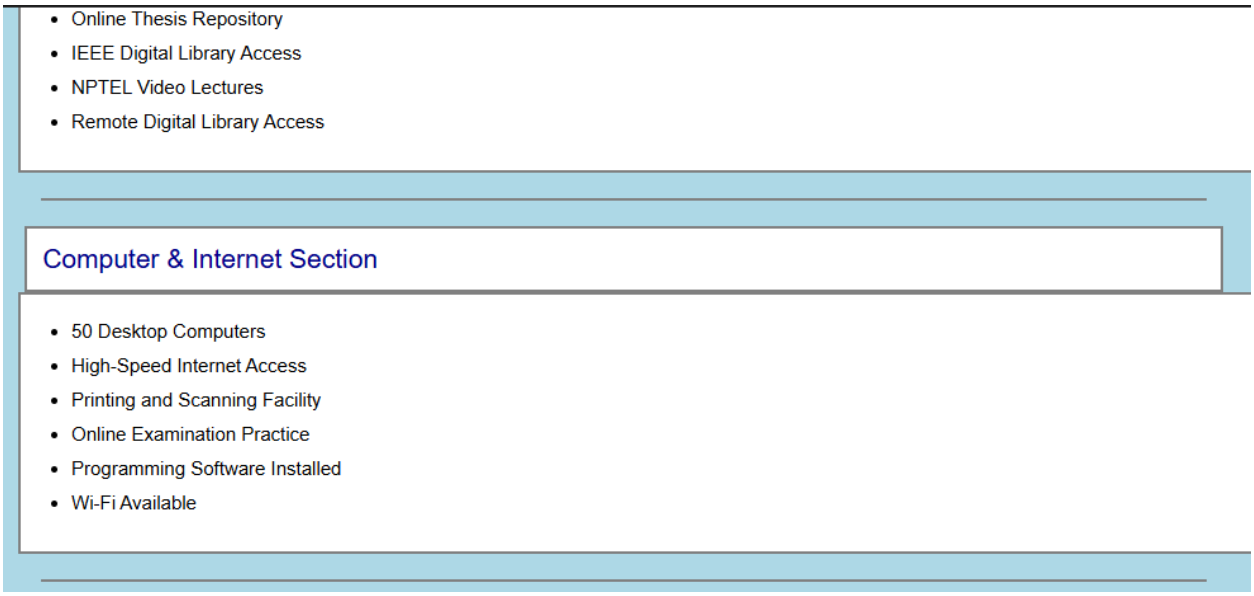
Payment Information

- Fees can be paid online through the StudentHub Portal.
- Students can pay semester-wise or annually.
- NRI fees are approximately double the regular tuition fee.
- Hostel, transport, examination and library fees are charged separately.
- Scholarships are available for eligible students according to university rules.

Back to Dashboard

Library :


StudentHub Library Portal
Library Information
Library Name: StudentHub Central Library
Location: Academic Block - A, Ground Floor
Working Days: Monday to Saturday
Library Timings: 8:00 AM - 8:00 PM
Digital Library Services
E-Books Collection
E-Journals
Research Papers
Previous Year Question Papers
Online Thesis Repository


Online Thesis Repository
IEEE Digital Library Access
NPTEL Video Lectures
Remote Digital Library Access
Computer & Internet Section
50 Desktop Computers
High-Speed Internet Access
Printing and Scanning Facility
Online Examination Practice
Programming Software Installed
Wi-Fi Available

ITUE:203-WDF **25DCE097**

Reading Hall

- Individual Reading Tables
- Silent Study Zone
- Air Conditioned Reading Room
- Newspaper & Magazine Section
- Open from 8:00 AM to 8:00 PM

Group Discussion & Project Area

- Project Discussion Rooms
- Group Study Tables
- Whiteboard Facility
- Project Presentation Area
- Meeting Rooms (Advance Booking Required)

Library Rules

1. Carry Student ID Card.
2. Maintain Silence.
3. No Food or Drinks Allowed.
4. Return Books Before Due Date.
5. Handle Books Carefully.
6. Mobile Phones Must Be Silent.

Library Contact

Librarian: Dr. Meera Shah

Email: library@studenthub.edu

Phone: +91 9876543210

Back to Dashboard

ITUE:203-WDF **25DCE097**

Notices :

StudentHub University

Official University Notice Board

Latest Announcements

- Academic Calendar for the current semester has been published.
- Online Examination Registration Portal is now open.
- Scholarship applications are available for eligible students.
- Library membership renewal has started.
- University Convocation registration is now open.
- Campus Placement Drive schedule has been updated.
- National Workshop on Artificial Intelligence will be held next month.
- Research paper submission guidelines have been updated.
- Annual Sports Festival registrations are now open.
- Cultural Festival event schedule has been published.

Upcoming Events

Date	Event
05 August 2026	Orientation Program
12 August 2026	Placement Training Session
18 August 2026	Research Workshop
25 August 2026	Annual Cultural Festival
05 September 2026	Industry Interaction Session
15 September 2026	Innovation & Startup Seminar

Important Links

- Academic Calendar
- Examination Schedule
- Scholarship Information
- Placement Cell Updates
- Library Announcements
- University Circulars
- Research Notices
- Student Activity Updates
- Holiday List
- Download Official Forms

Back to Dashboard

About Us :

StudentHub University

About Us

Our Vision

StudentHub University aims to become a globally recognized institution by providing quality education, encouraging innovation, promoting research, and preparing students to become responsible professionals and future leaders.

Our Mission

- Provide quality education through experienced faculty.
- Promote research, innovation and entrepreneurship.
- Develop leadership and communication skills.
- Strengthen collaboration with industries and organizations.
- Encourage ethical values and social responsibility.
- Create lifelong learning opportunities for students.

ITUE:203-WDF **25DCE097**

Academic Programs

StudentHub University offers undergraduate, postgraduate and doctoral programs in various disciplines to prepare students for successful careers.

- Engineering & Technology
- Computer Applications
- Management Studies
- Commerce
- Science
- Health Sciences
- Arts & Humanities
- Pharmacy

Explore More

To know more about placements, campus facilities, sports, research, student clubs, cultural activities, international collaborations and student support services, visit the Campus Life page.

Campus Life & Student Opportunities

Back to Dashboard

ITUE:203-WDF **25DCE097**

Campus Life :

StudentHub University

Official University Notice Board

Stay updated with the latest university announcements, examination notices, placement drives, scholarships, events, and important academic information.

Latest Announcements

- Academic Calendar for Semester 2026-27 has been published.
- Online Examination Registration Portal is now open.
- Library Membership Renewal has started.
- Scholarship Applications are available for eligible students.
- University Convocation Registration is now open.
- Campus Placement Drive schedule has been updated.
- National Workshop on Artificial Intelligence will be conducted next month.
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Upcoming Events

Date	Event
05 August 2026	Orientation Program for New Students
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18 August 2026	Research & Innovation Workshop
25 August 2026	Annual Cultural Festival
05 September 2026	Industry Interaction Session
15 September 2026	Startup & Entrepreneurship Seminar

ITUE:203-WDF **25DCE097**

Examination Notices

Notice	Status
Semester Examination Time Table	Available
Online Exam Registration	Open
Hall Ticket Download	Available Soon
Rechecking & Reassessment Forms	Closed
Result Declaration	To Be Announced

Admission Updates

- Admissions for Academic Year 2026-27 are Open.
- Merit List will be published on 10 August 2026.
- Document Verification Schedule Released.
- Hostel Admission Forms Available.
- Fee Payment Last Date: 20 August 2026.

Placement Cell Updates

Company	Drive Date
TCS	22 August 2026
Infosys	28 August 2026
Wipro	05 September 2026
LTIMindtree	10 September 2026
HCL Technologies	18 September 2026

Scholarships & Financial Aid

- Government Scholarship Applications Open.
- Merit Scholarships for University Rank Holders.
- Sports Scholarships Available.
- Financial Assistance for Economically Weaker Students.
- Last Date for Scholarship Submission: 30 August 2026.

ITUE:203-WDF **25DCE097**

Library Updates

- Digital Library Portal Updated.
- New Engineering, Medical and Commerce Books Added.
- Extended Reading Hall Timings During Examination.
- Research Journal Access Available.
- Library Open from 8:00 AM to 10:00 PM.

Quick Information

Service	Status
Academic Calendar	Available
Examination Schedule	Available
Library Portal	Open
Fee Payment Portal	Open
Student Help Desk	Monday - Saturday

Important Links

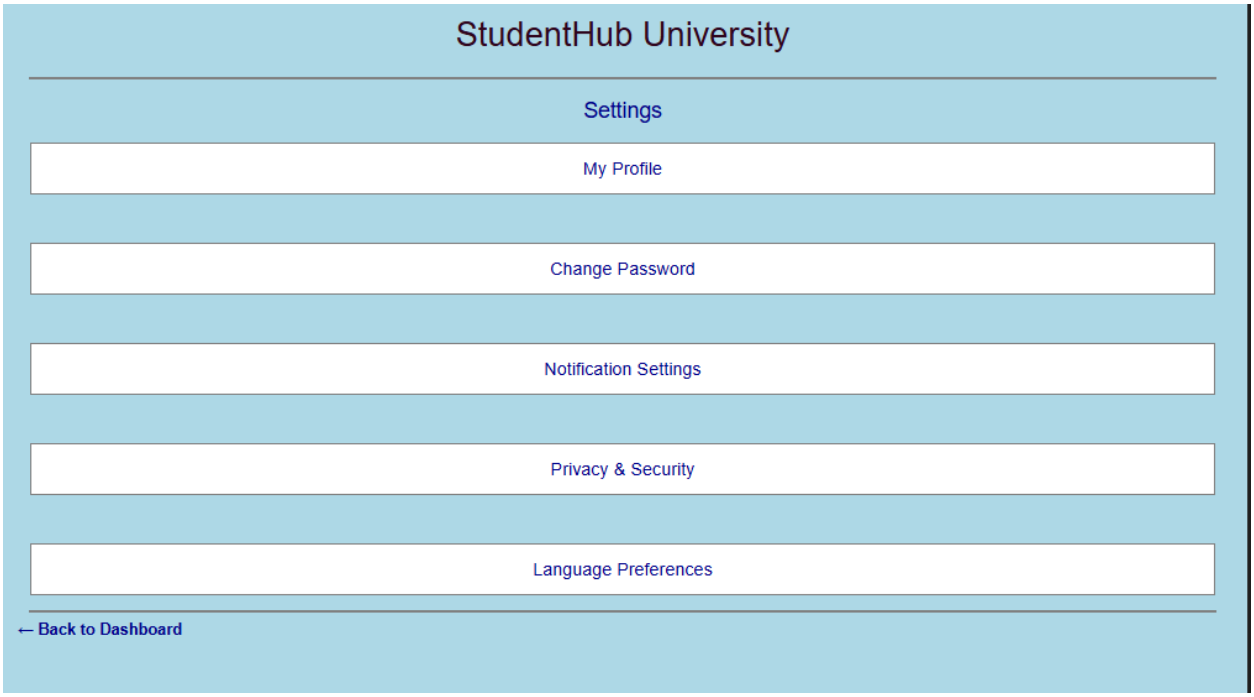
- Academic Calendar
- Examination Schedule
- Scholarship Information
- Placement Cell Updates
- Library Announcements
- Research Notices
- Holiday List
- Download Official Forms

Back to Dashboard

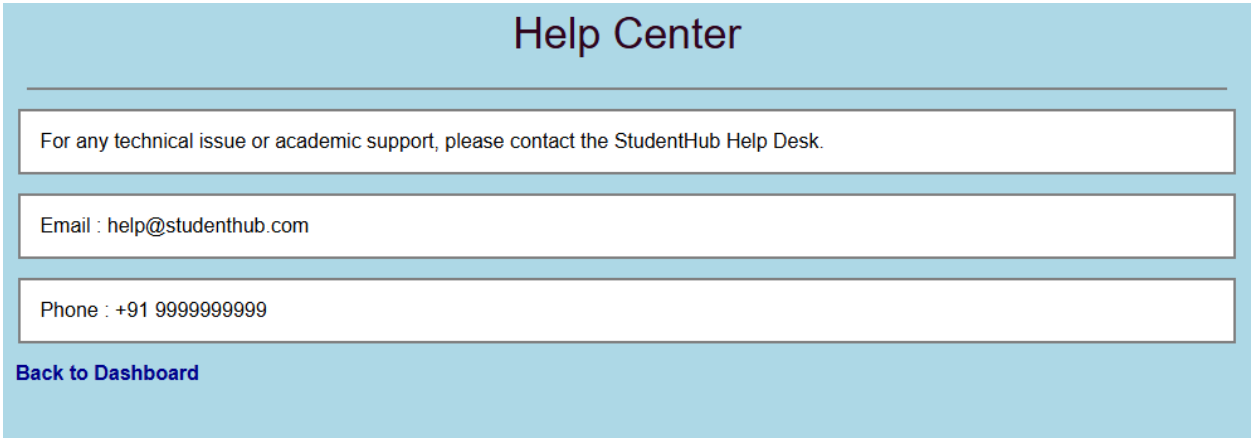
Contact Us :



Settings :



Help :



ITUE:203-WDF **25DCE097**

Learning Outcomes:

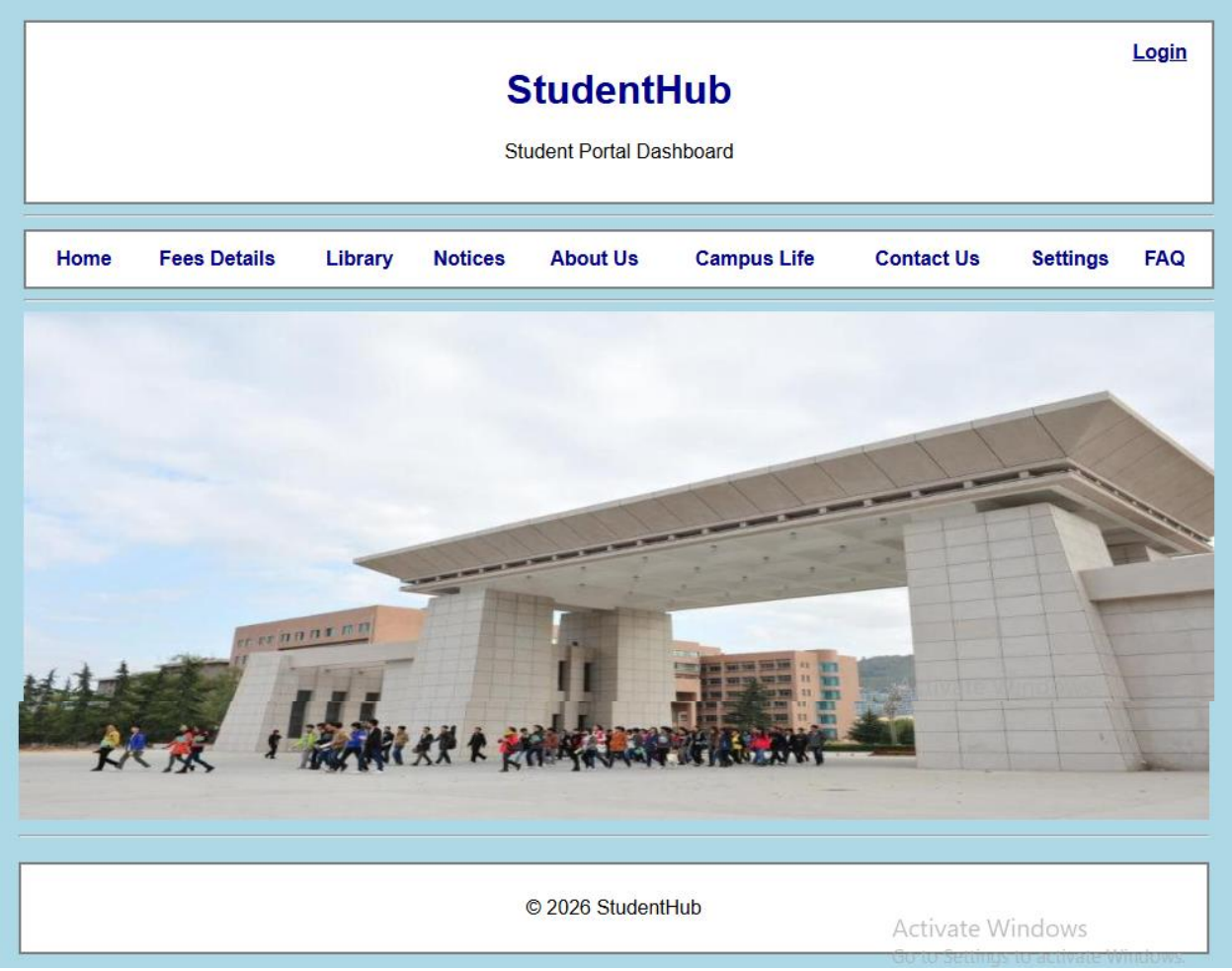
After completing this practical, students will be able to understand responsive web design and use CSS Grid and Flexbox to create flexible layouts. They will learn how to make webpages suitable for mobile, tablet, and desktop screens. Students will also understand the importance of proper typography, spacing, colors, and contrast and learn to use CSS frameworks consistently without unnecessary inline CSS.

Conclusion:

This practical helped in understanding how responsive layouts are created using CSS Grid, Flexbox, and CSS frameworks. The pages were designed to adjust according to different screen sizes while maintaining proper alignment, spacing, and readability. Responsive design makes the StudentHub Portal more user-friendly and accessible on different devices.

Changes That I made In my webpages :

This Is My Final dashboard picture at first I have A dashboard which was not looking this much Identical So I change my Dashboard design with this.



Also I Have changed my other pages In all the pages there is option get back to dashboard which I change with the arrow at top left in each page. Like this :

←

StudentHub University

Academic Fee Structure (Sample)

Academic Year : 2026-2027

Engineering Programs

Course	Annual Tuition Fee	NRI Fee
B.Tech Computer Engineering	₹1,45,000	₹2,90,000
B.Tech Information Technology	₹1,45,000	₹2,90,000
B.Tech Artificial Intelligence & Data Science	₹1,45,000	₹2,90,000
B.Tech Mechanical Engineering	₹1,45,000	₹2,90,000
B.Tech Civil Engineering	₹1,45,000	₹2,90,000
B.Tech Electrical Engineering	₹1,45,000	₹2,90,000
B.Tech Electronics & Communication	₹1,45,000	₹2,90,000

Activate Windows
Go to Settings to activate Windows.

←

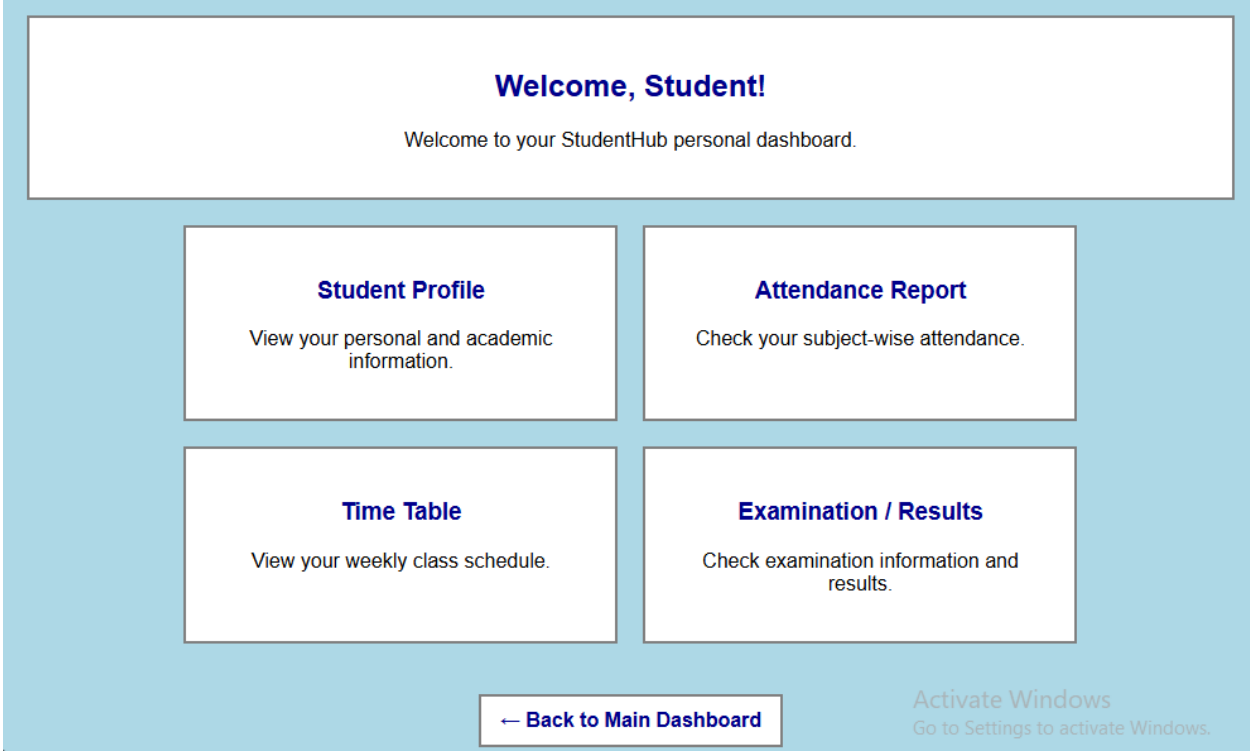
Contact Us

Email : studenthub@gmail.com

Phone : +91 9876543210

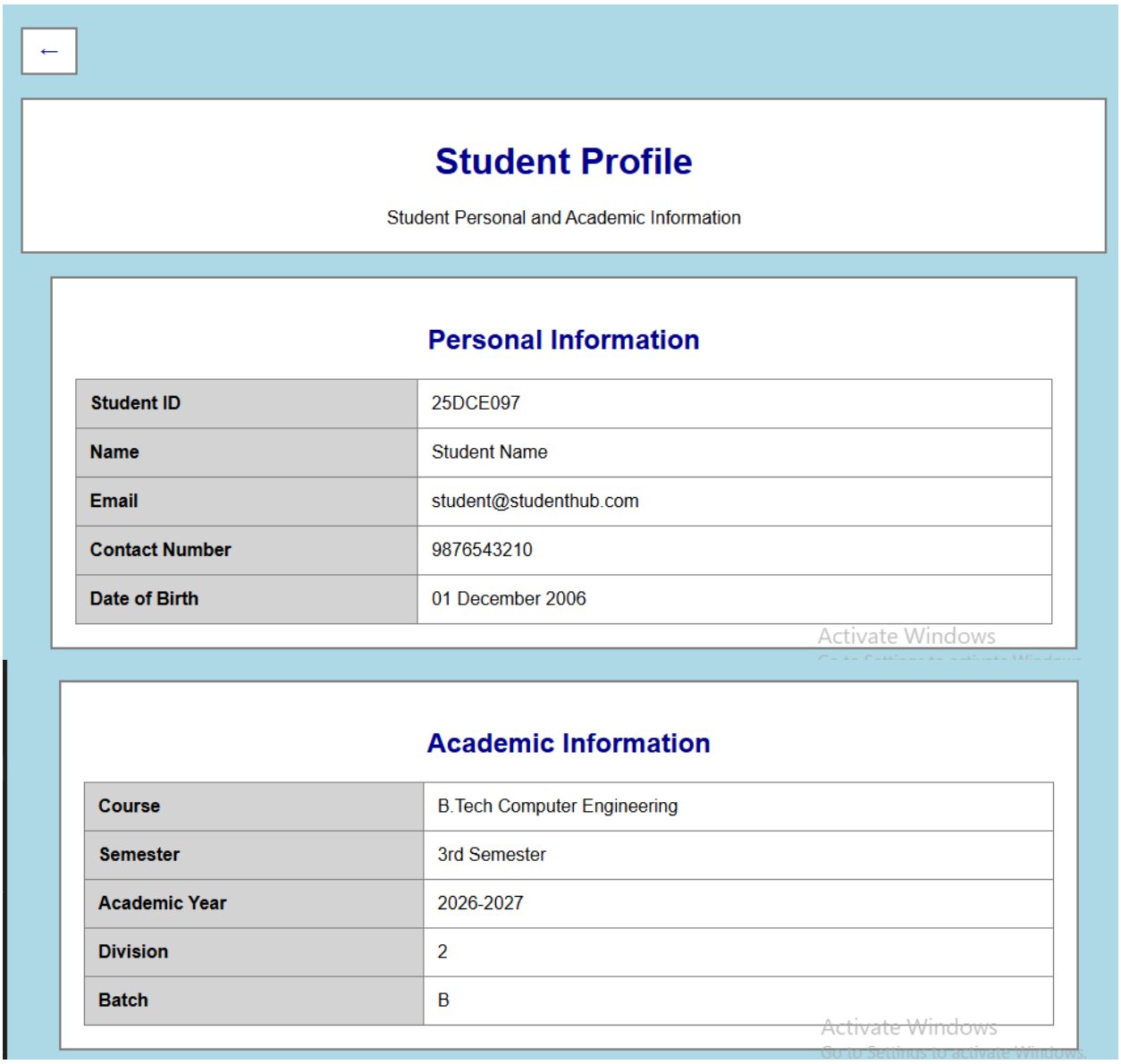
Address : CHARUSAT University, Changa, Gujarat

Also I add a second dashboard for the students who already studying in that University .That Second Dashboard looks Like this :



Student Profile :

ITUE:203-WDF **25DCE097**



←

Student Profile

Student Personal and Academic Information

Personal Information

Student ID	25DCE097
Name	Student Name
Email	student@studenthub.com
Contact Number	9876543210
Date of Birth	01 December 2006

Activate Windows

Academic Information

Course	B.Tech Computer Engineering
Semester	3rd Semester
Academic Year	2026-2027
Division	2
Batch	B

Activate Windows
Go to Settings to activate Windows.

Attandance Report :

ITUE:203-WDF **25DCE097**

←

Attendance Report

Academic Year: 2026-2027

Subject	Total Classes	Classes Attended	Attendance
Object Oriented Programming	40	36	90%
Computer Networks	38	33	87%
Database Management	42	37	88%
Web Development	40	34	85%
Software Engineering	36	32	89%

Overall Attendance

Your overall attendance is **88%**.

Activate Windows
Go to Settings to activate Windows.

Time Table :

←

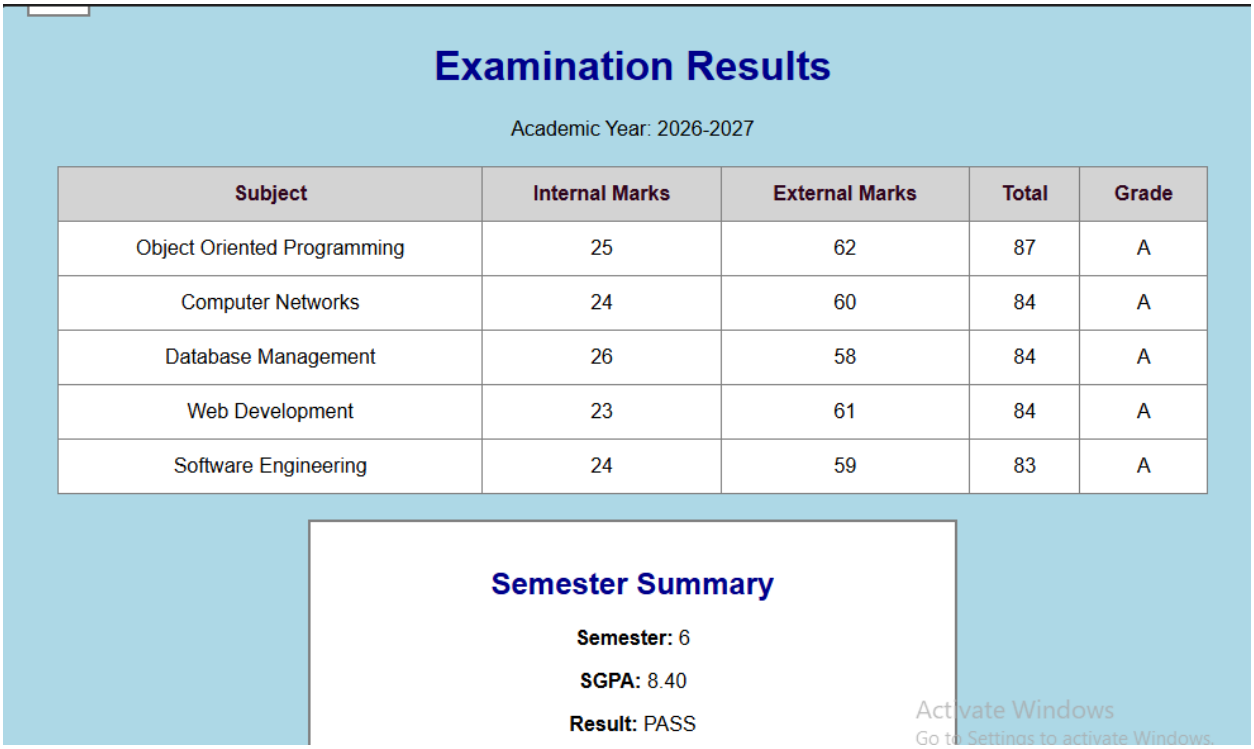
Class Time Table

Academic Year: 2026-2027

Day	9:00 - 10:00	10:00 - 11:00	11:00 - 12:00	12:00 - 1:00	2:00 - 3:00	3:00 - 4:00
Monday	OOP	Computer Networks	DBMS	Break	Web Development	Software Engineering
Tuesday	DBMS	OOP	Web Development	Break	Computer Networks	Lab
Wednesday	Software Engineering	DBMS	Computer Networks	Break	OOP Lab	OOP Lab
Thursday	Web Development	Computer Networks	OOP	Break	DBMS Lab	DBMS Lab
Friday	Computer Networks	Software Engineering	DBMS	Break	Web Development	Library

Activate Windows
Go to Settings to activate Windows.

Results :



Examination Results

Academic Year: 2026-2027

Subject	Internal Marks	External Marks	Total	Grade
Object Oriented Programming	25	62	87	A
Computer Networks	24	60	84	A
Database Management	26	58	84	A
Web Development	23	61	84	A
Software Engineering	24	59	83	A

Semester Summary

Semester: 6

SGPA: 8.40

Result: PASS

Activate Windows
Go to Settings to activate Windows.

Practical 4: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition

To implement dynamic user interface components such as a collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript, HTML5, and CSS3. The objective is to improve user experience through interactive and responsive frontend behaviour.

Questions and Solutions

Q1. How are DOM elements selected and modified?

Solution: DOM elements are selected using methods such as `getElementById()`, `querySelector()`, and `querySelectorAll()`. Their content, styles, attributes, and classes can be modified using JavaScript.

Q2. Are event listeners attached correctly?

Solution: Event listeners are attached using addEventListener(). They respond to user actions such as clicks, button presses, and menu selections to perform the required operations.

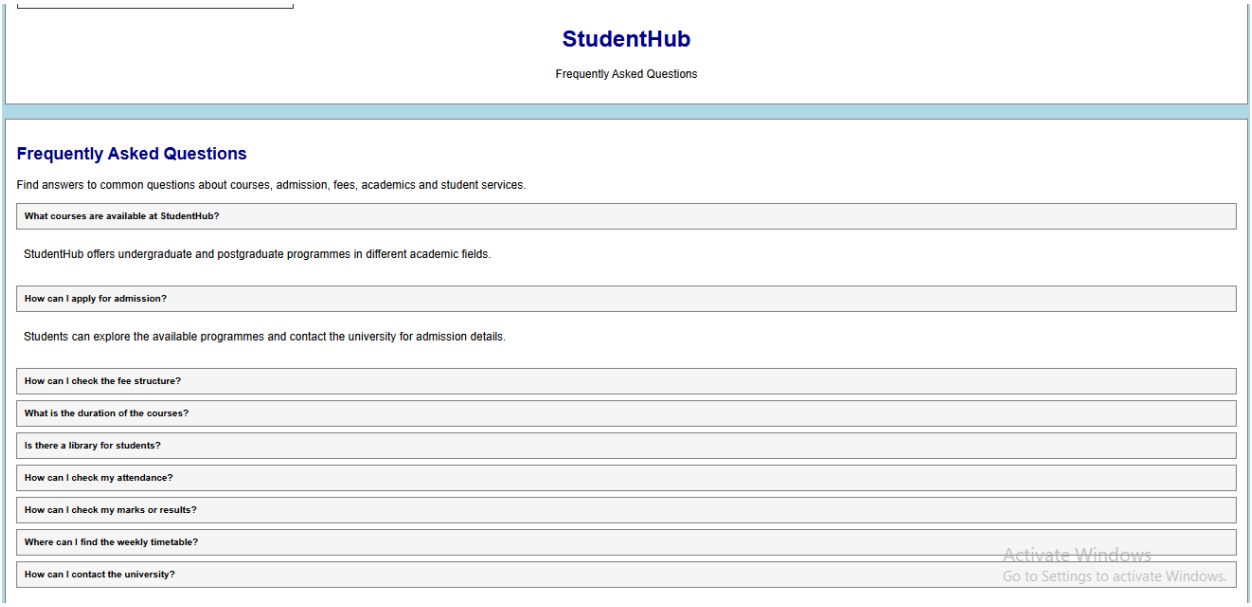
Q3. Is localStorage used for remembering theme choice?

Solution: Yes, localStorage stores the user's selected theme in the browser. The saved theme is retrieved when the page reloads, allowing the website to maintain the user's preference.

Q4. Does interactivity improve usability without breaking accessibility?

Solution: Yes. Interactive components improve navigation and usability when implemented correctly. Keyboard accessibility, appropriate labels, visible focus indicators, and proper handling of modal popups help ensure accessibility.

Output :



3. Learning Outcomes

1. Understand the Document Object Model (DOM) and its role in webpage manipulation.
2. Implement JavaScript event handling to respond to user interactions.
3. Create interactive UI components such as FAQs, modals, sliders, notification banners, and hamburger menus.
4. Use localStorage to save and restore user interface preferences.

5. Apply CSS classes and transition effects to improve the appearance and usability of a website.
6. Test JavaScript functionality and identify errors using browser developer tools.

4. Conclusion

In this practical, JavaScript was used to manipulate DOM elements, handle user events, and develop interactive user interface components. Features such as collapsible FAQs, modal popups, sliders, notification banners, hamburger menus, and theme switching improved the website's functionality and user experience. The use of localStorage helped preserve user preferences across page reloads. This practical demonstrated how HTML, CSS, and JavaScript work together to create dynamic and user-friendly web applications.

Is this conversation helpful so far?

Practical 5

Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition : Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password,course, year, gender, and terms acceptance. Use Regular Expression for validations.

- Create a user-friendly registration form.
- Validate user input using JavaScript.
- Display clear error messages.
- Check password strength.
- Make the form accessible and easy to use.

Questions:

1. Are correct input types and attributes used?

Yes. Appropriate HTML5 input types and attributes are used.

Examples:

- type="text" for name

- type="email" for email
- type="tel" for mobile
- type="password" for passwords
- required to make important fields compulsory
- <select> for course and year
- Radio buttons for gender
- Checkbox for terms acceptance

2. Are validation errors displayed near the relevant fields?

Yes. Error messages are displayed immediately below the corresponding input field.

Example:

Email
[abc.com]

Enter a valid email address.

This makes it easy for the user to identify and correct the mistake.

3. Is password strength checked?

Yes. Password strength is checked using JavaScript.

The password is checked for:

- Minimum 8 characters
- Uppercase letter
- Number
- Special character

The password is then shown as:

Weak → Medium → Strong

A password-strength bar is also displayed.

4. Is the form accessible using keyboard and screen-reader-friendly labels?

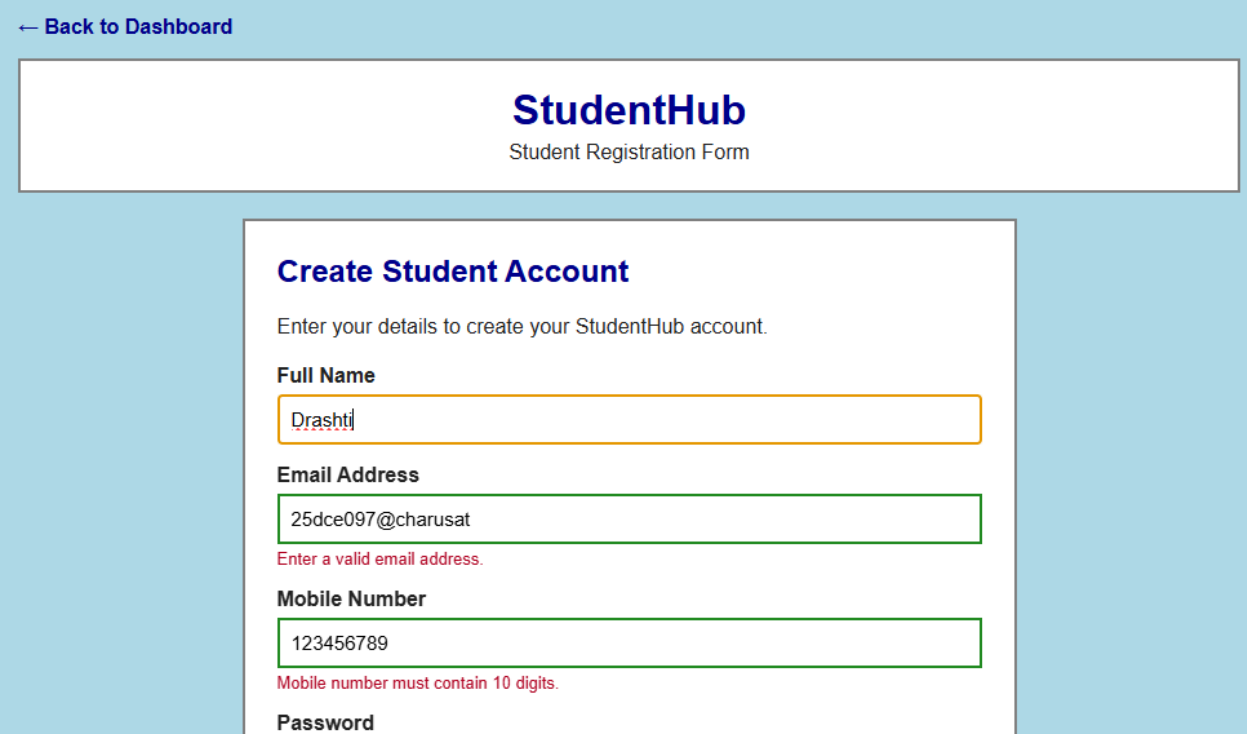
Yes. Each input has a proper <label> connected using the for attribute.

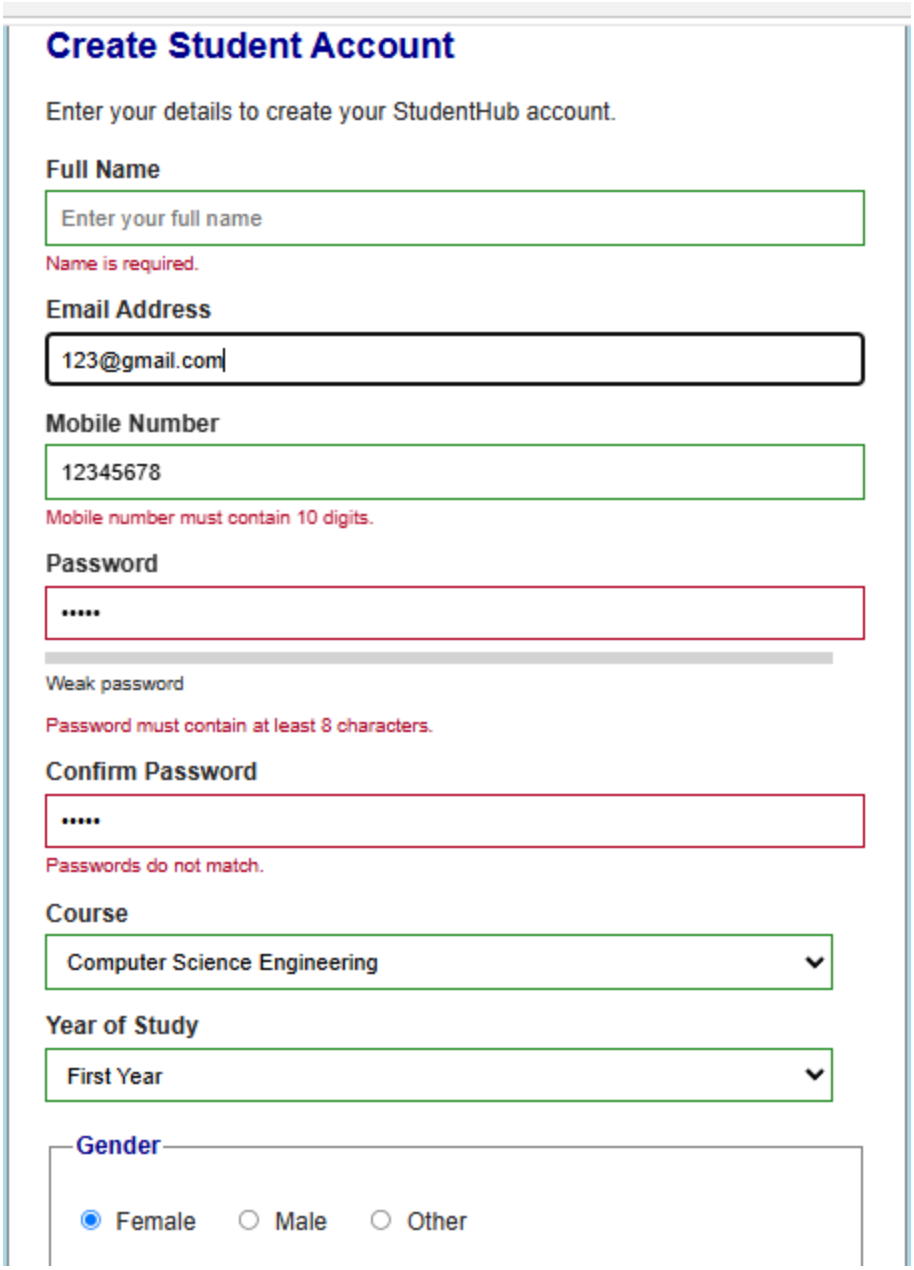
Example:

```
<label for="email">Email Address</label>

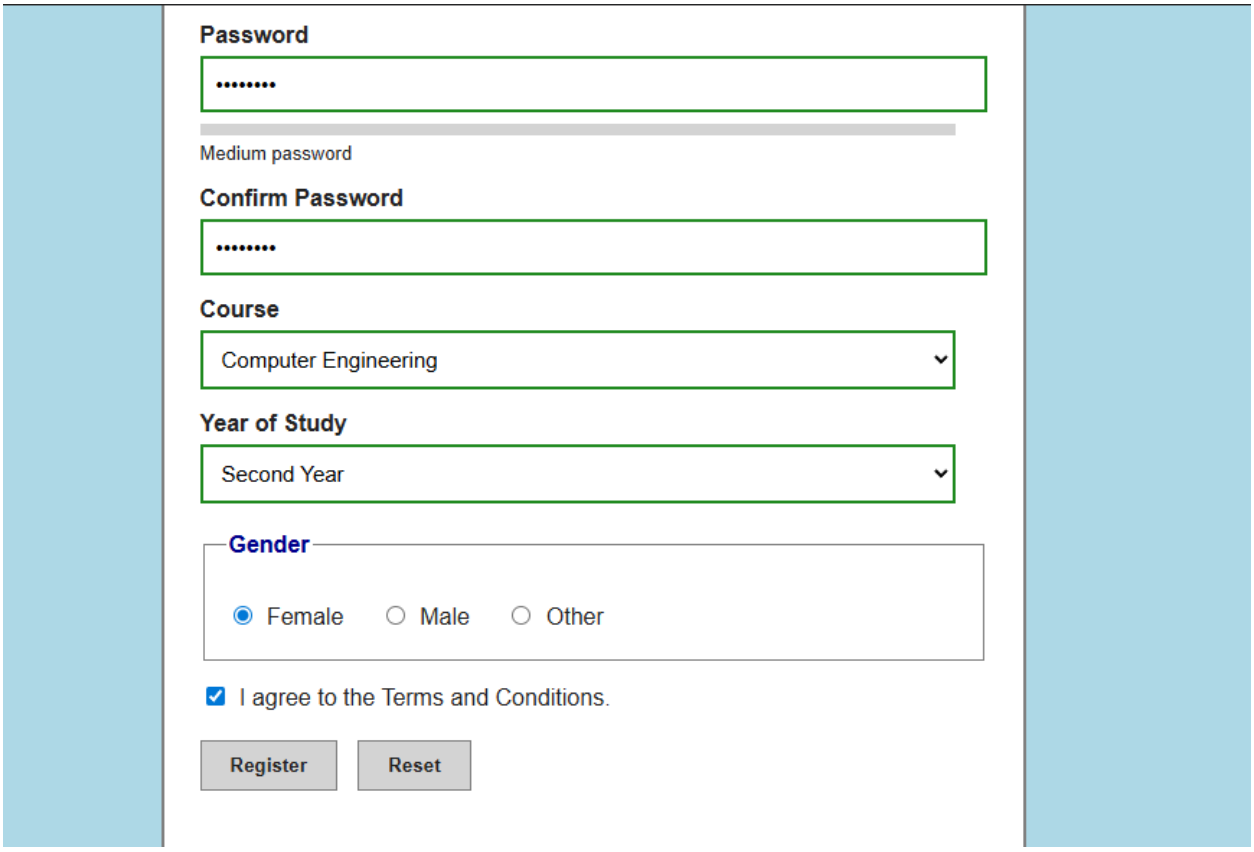
<input type="email" id="email">
```

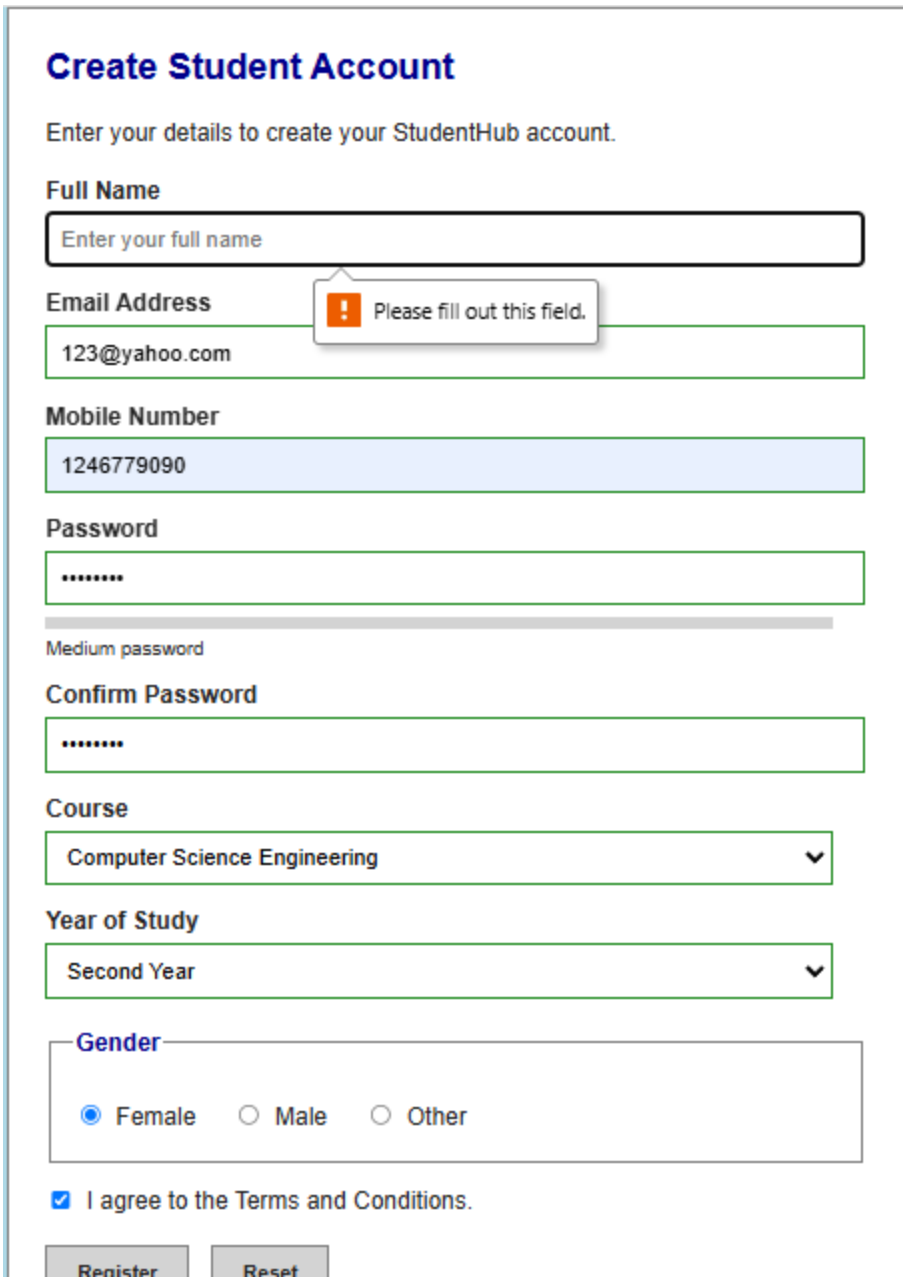
This helps **screen readers** identify the purpose of each field. Standard form controls also allow users to navigate the form using the **keyboard**.




Create Student Account
Enter your details to create your StudentHub account.
Full Name
Enter your full name
Name is required.
Email Address
123@gmail.com
Mobile Number
12345678
Mobile number must contain 10 digits.
Password
•••••
Weak password
Password must contain at least 8 characters.
Confirm Password
•••••
Passwords do not match.
Course
Computer Science Engineering
Year of Study
First Year
Gender
Female
Male
Other

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Conclusion :

Practical 5 demonstrates how HTML5 and JavaScript can be used together to create a secure, user-friendly, validated student registration form with Regular Expressions, password-strength checking, clear error handling, and accessibility.

Practical 6: Rendering External JSON Data using Fetch API, Search and Filter

Problem Definition

To fetch external JSON data using the Fetch API and display it dynamically on a webpage. Implement search, filtering, sorting, pagination, and proper loading and error handling.

Key Questions and Answers

1. How is JSON fetched, parsed, and rendered?

JSON data is fetched from an external file using the fetch() method. The response is converted into JavaScript objects using the .json() method. JavaScript then renders the data dynamically on the webpage by creating HTML elements and displaying the required information.

2. Which array methods are used for search, filter, sort, and pagination?

The filter() method selects records matching a search term or condition. The sort() method arranges records in ascending or descending order. The slice() method selects a limited number of records for pagination, while map() can be used to generate HTML content from array data.

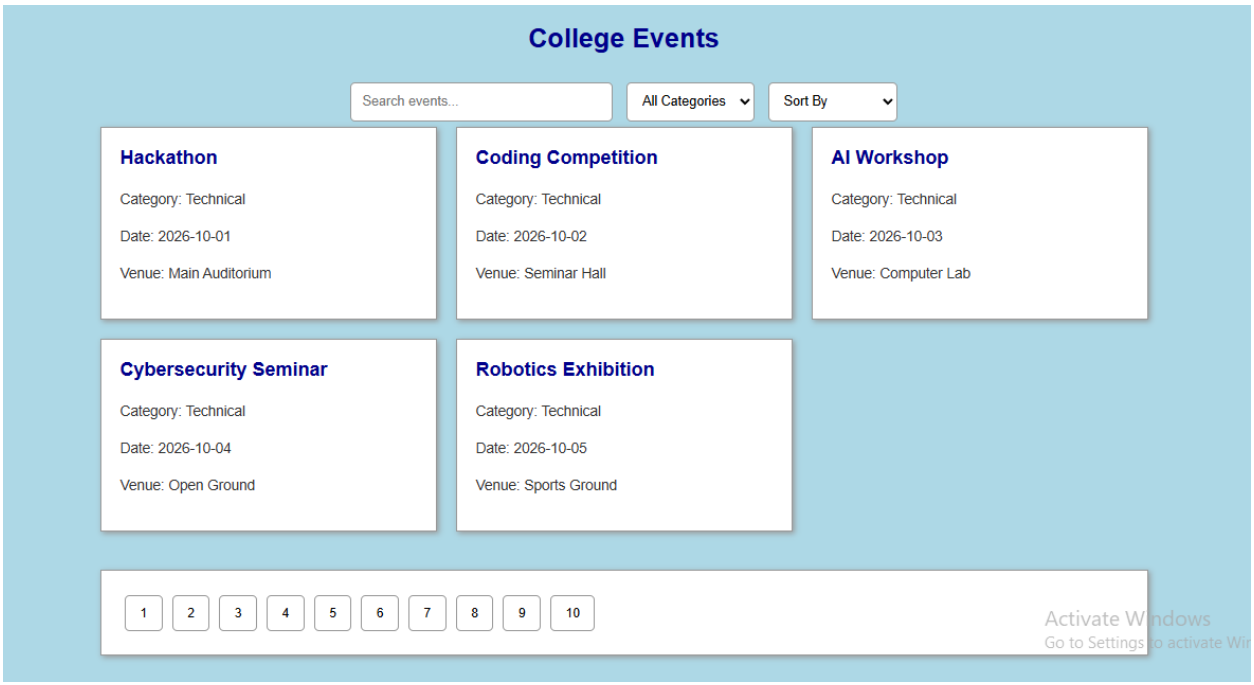
3. How are loading and error states handled?

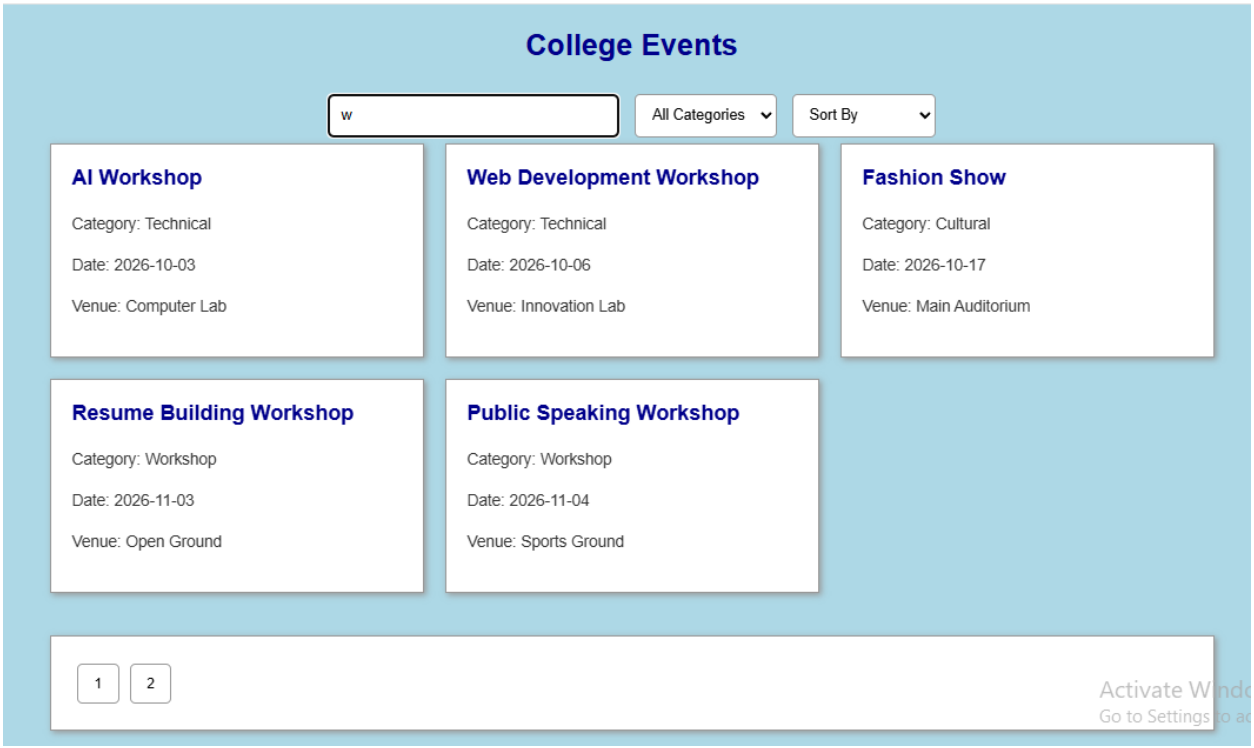
A loading message is displayed while the JSON data is being fetched. If the request fails or the file cannot be loaded, an error message is displayed using .catch() or try...catch. The HTTP response status should also be checked to identify unsuccessful requests.

4. How is modularity maintained in JavaScript files?

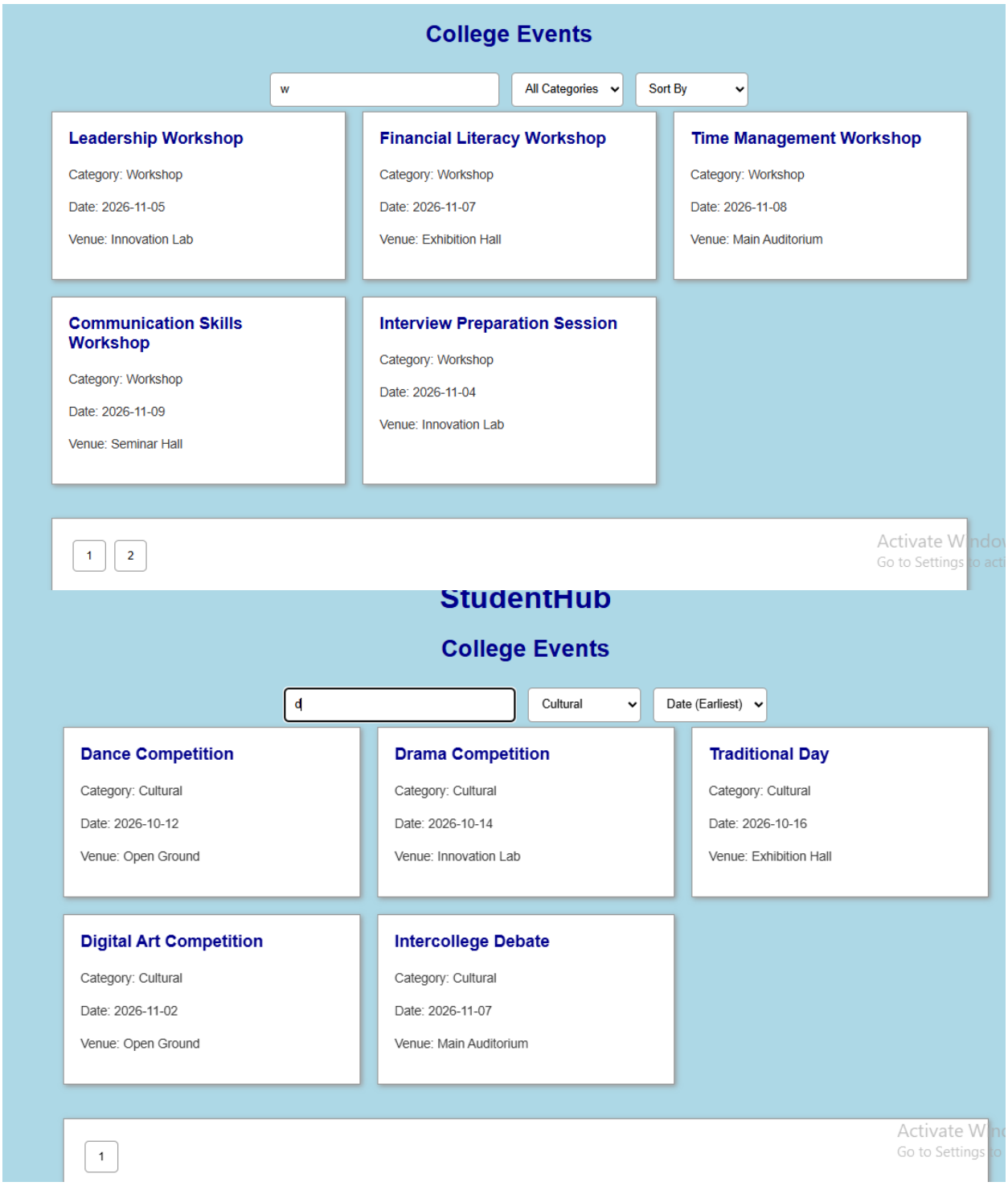
Modularity is maintained by dividing the code into separate functions or files. For example, one function fetches JSON data, another renders the records, and other functions handle searching, filtering, sorting, and pagination. This makes the code easier to understand, test, and maintain.

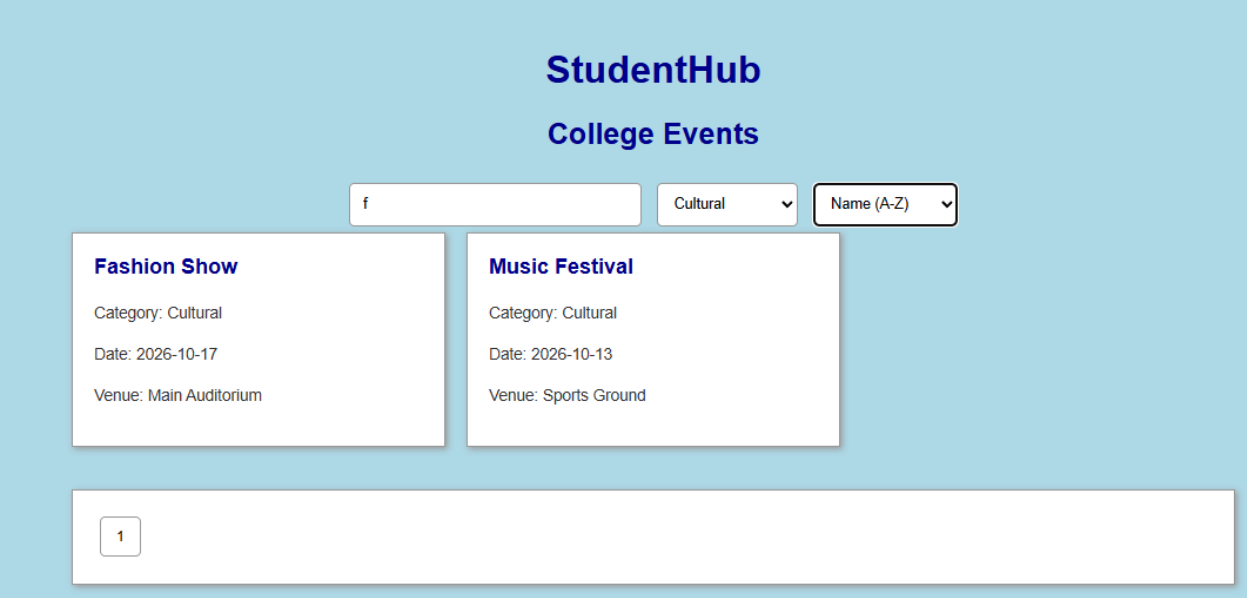
Output :


College Events
Search events...
All Categories
Sort By
Hackathon
Category: Technical
Date: 2026-10-01
Venue: Main Auditorium
Coding Competition
Category: Technical
Date: 2026-10-02
Venue: Seminar Hall
AI Workshop
Category: Technical
Date: 2026-10-03
Venue: Computer Lab
Cybersecurity Seminar
Category: Technical
Date: 2026-10-04
Venue: Open Ground
Robotics Exhibition
Category: Technical
Date: 2026-10-05
Venue: Sports Ground
1 2 3 4 5 6 7 8 9 10
Activate Windows


College Events
w
All Categories
Sort By
AI Workshop
Category: Technical
Date: 2026-10-03
Venue: Computer Lab
Web Development Workshop
Category: Technical
Date: 2026-10-06
Venue: Innovation Lab
Fashion Show
Category: Cultural
Date: 2026-10-17
Venue: Main Auditorium
Resume Building Workshop
Category: Workshop
Date: 2026-11-03
Venue: Open Ground
Public Speaking Workshop
Category: Workshop
Date: 2026-11-04
Venue: Sports Ground
1 2
Activate Windo

ITUE:203-WDF **25DCE097**





Conclusion

The Fetch API was used to retrieve JSON data and display it dynamically. Search, filtering, sorting, pagination, and error handling made the webpage more interactive and user-friendly.

Practical 7: PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

Problem Definition

To develop a registration or contact form using PHP that processes submitted data through the POST method. Validate and sanitize user inputs on the server side, store valid records in CSV or JSON files, and display appropriate success or error messages. This practical demonstrates basic server-side form processing and file handling.

Key Questions / Analysis / Interpretation

1. Is the form submitted using POST?

Yes, the form uses the POST method to send user data to the PHP server. Unlike the GET method, POST does not display submitted form values directly in the URL. In PHP, submitted values can be accessed using the $_POST superglobal array.

2. Are inputs validated and sanitized server side?

Yes, server-side validation checks whether the submitted data meets the required conditions, such as valid email format, required fields, and correct input length. Sanitization helps remove or handle unwanted characters. These checks are performed in PHP because client-side validation alone can be bypassed.

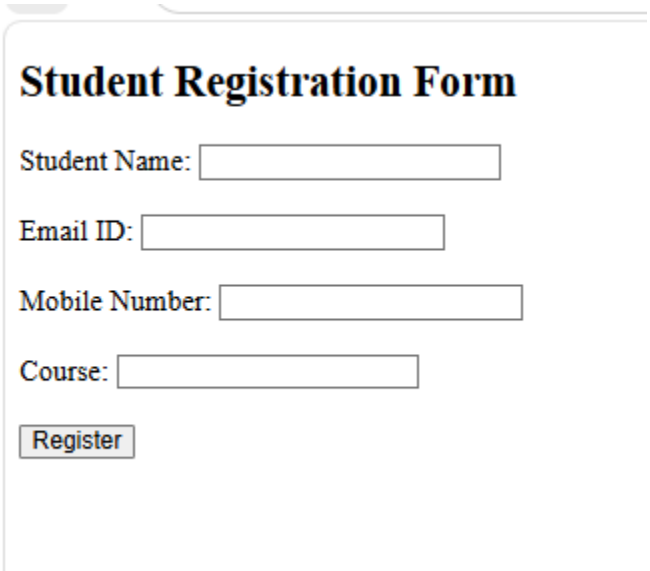
3. Is file writing handled safely?

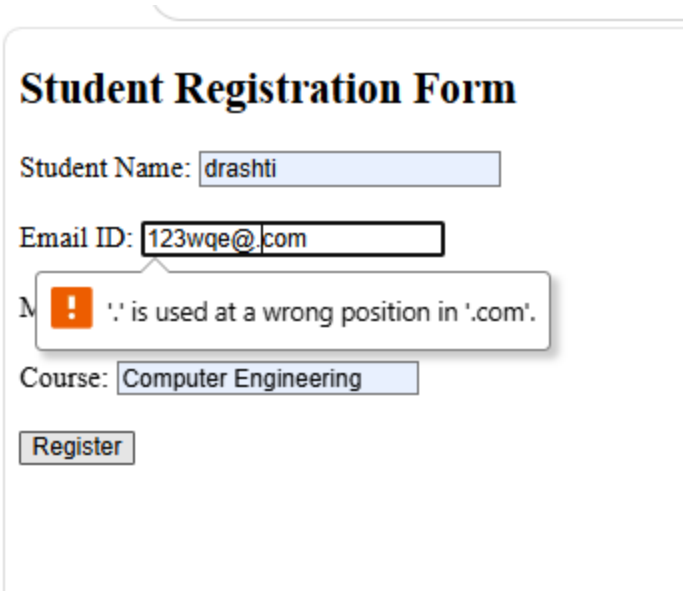
Yes, file writing should include proper error handling and appropriate file permissions. Functions such as fopen(), fputcsv(), and fclose() can be used for CSV storage, while file_put_contents() and json_encode() can be used for JSON storage. File locking and input validation help reduce the risk of data corruption and unsafe file handling.

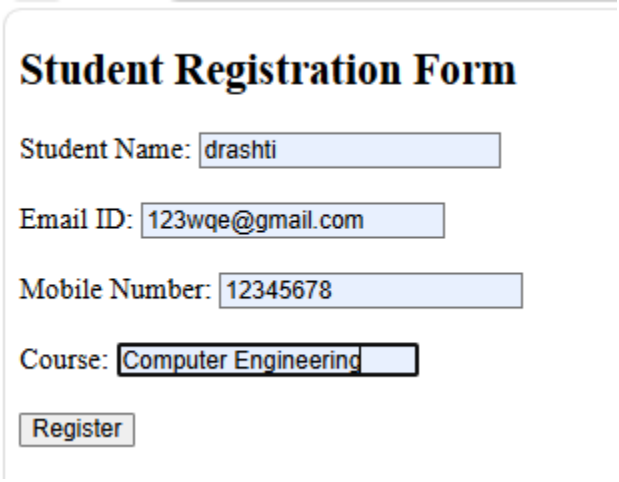
4. Are success and error responses displayed clearly?

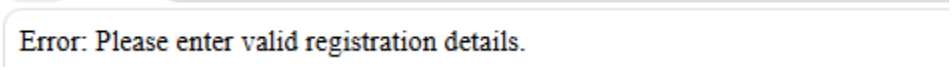
Yes, PHP displays a success message after valid data has been stored successfully. If required fields are missing, the input is invalid, or file storage fails, an appropriate error message is displayed. User-provided data should be escaped before being displayed on a webpage.

Output:




Student Registration Form
Student Name: drashti
Email ID: 123wqe@.com
'.' is used at a wrong position in '.com'.
Course: Computer Engineering
Register


Student Registration Form
Student Name: drashti
Email ID: 123wqe@gmail.com
Mobile Number: 12345678
Course: Computer Engineering
Register


Error: Please enter valid registration details.

Student Registration Form

Student Name: drashti

Email ID: 123wqe@gmail.com

Mobile Number: 1234567890

Course:

Regis Please fill out this field.

Registration Successful!

Name: drashti

Email: 123wqe@gmail.com

Mobile: 1234567890

Course: Computer Engineering

Student Registration Form

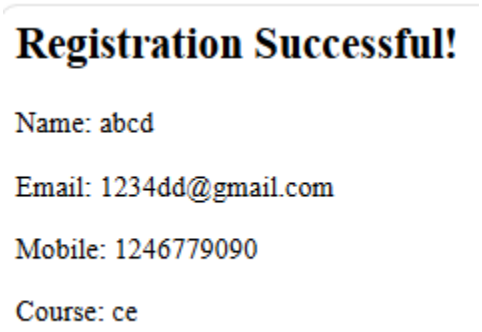
Student Name: abcd

Email ID: 1234dd@gmail.com

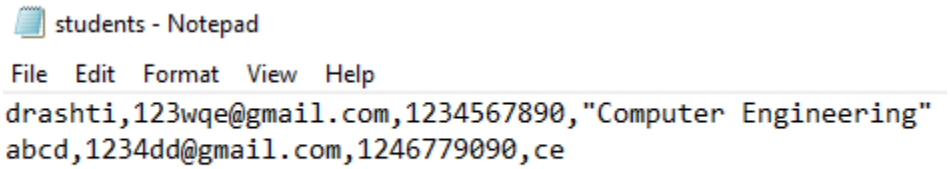
Mobile Number: 1246779090

Course: ce

Register



CSV file



Conclusion

In this practical, a PHP form was used to collect and process user information through the POST method. Server-side validation and sanitization helped ensure that the submitted data was valid, while CSV or JSON file storage allowed records to be saved. Success and error messages made the form easier to use and understand.

Practical 8: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition

To design a normalized MySQL database for the StudentHub application containing students, events, and registrations. Create the tables with appropriate primary and foreign keys, represent their relationships using an Entity-Relationship (ER) model, and establish a PHP database connection. Use PDO and prepared statements to securely interact with the database.

Key Questions / Analysis / Interpretation

1. Is the schema normalized properly?

Yes, database normalization organizes information into related tables and reduces unnecessary data duplication. For example, student details are stored in the students table, event details in the events table, and registration information in the registrations table. This structure improves consistency and makes data easier to manage.

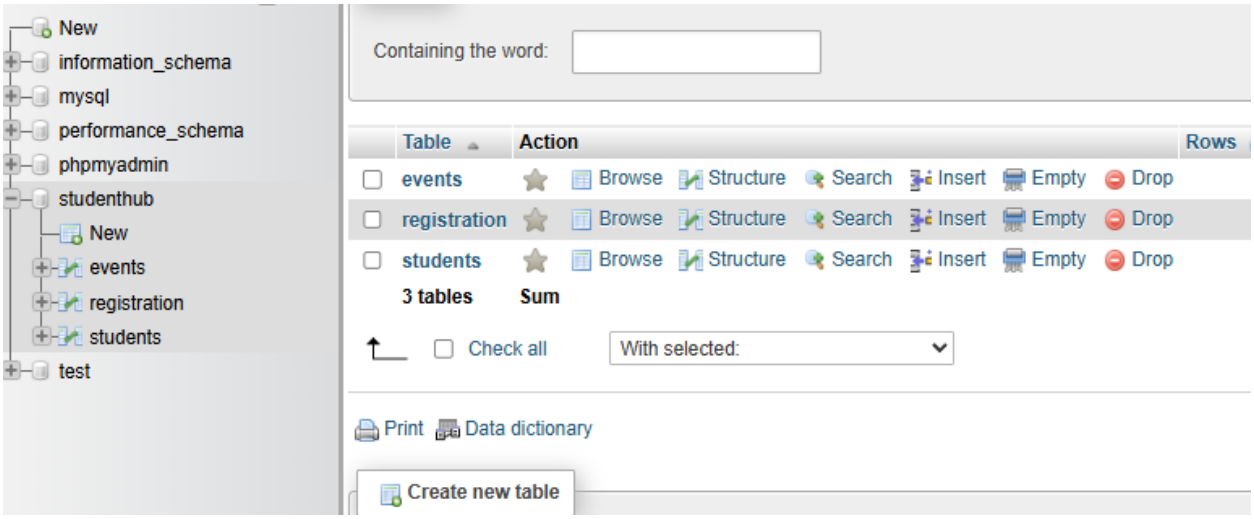
2. Are primary keys and foreign keys correctly defined?

Primary keys uniquely identify each record in a table. For example, student_id identifies a student, while event_id identifies an event. Foreign keys establish relationships between tables. In the registrations table, student_id and event_id can reference the corresponding records in the students and events tables, maintaining referential integrity.

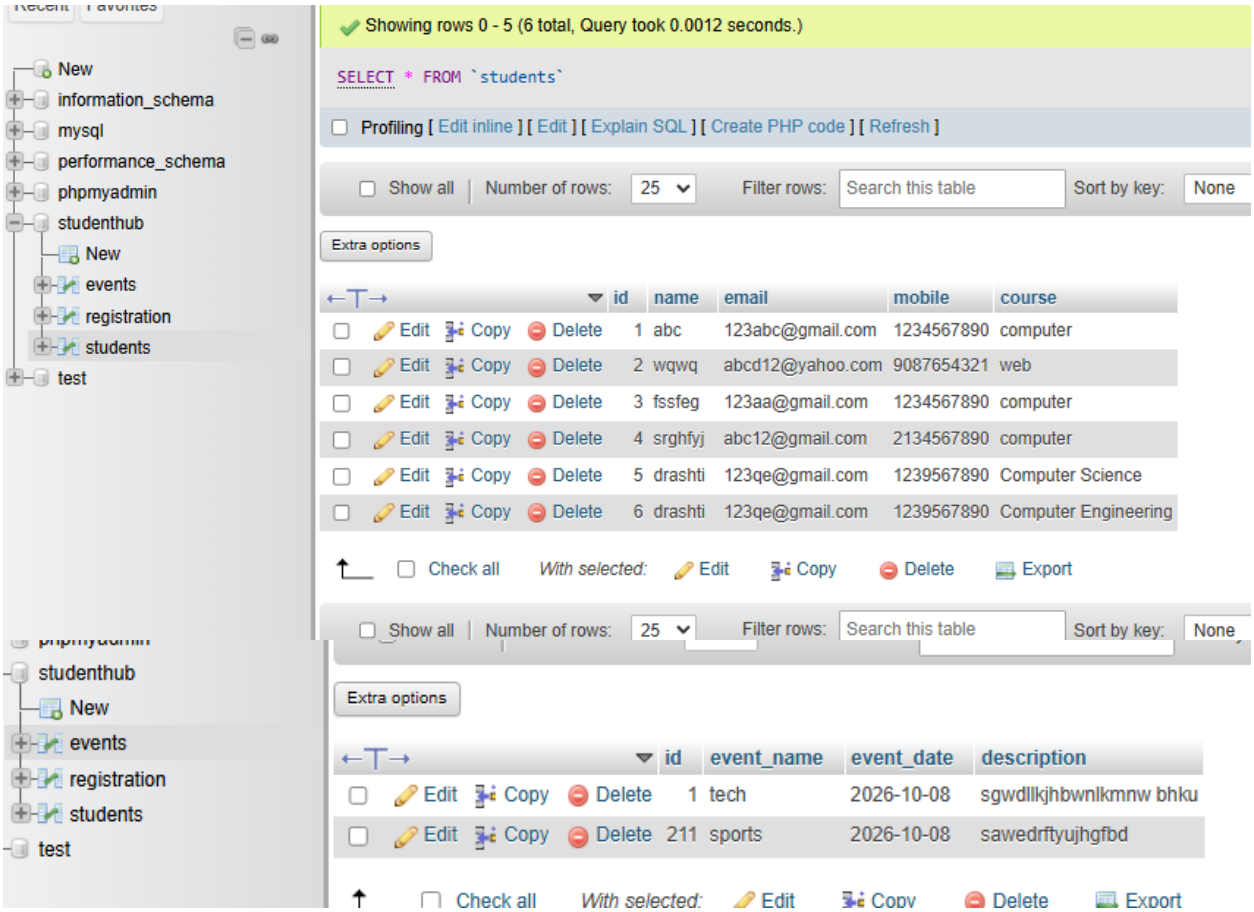
3. Is the database connection handled securely?

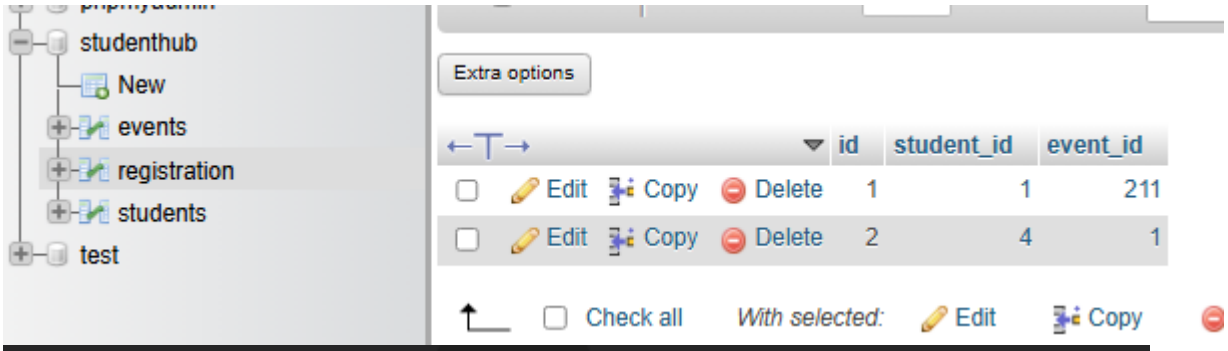
Yes, PHP can connect to MySQL using PDO with proper database credentials and exception-based error handling. Credentials should be stored securely rather than exposed publicly. Prepared statements should be used for queries containing user input to help prevent SQL injection attacks.

Output:



ITUE:203-WDF **25DCE097**





ister.php process.php db.php

lb.php

```php
<?php

$host = "localhost";
$dbname = "studenthub";
$username = "root";
$password = "";

try {
    $conn = new PDO(
        "mysql:host=$host;dbname=$dbname",
        $username,
        $password
    );

    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

} catch (PDOException $e) {
    echo "Connection failed: " . $e->getMessage();
}

?>
```

Conclusion

In this practical, a normalized MySQL database was designed for StudentHub using students, events, and registrations tables. Primary keys and foreign keys established relationships between the tables, while PDO provided connectivity between PHP and MySQL. Prepared statements and secure credential handling helped improve database security and reliability.

Practical 9: Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing

1. Aim

To develop a secure user registration system using PHP and MySQLi with frontend and backend validation, duplicate email and username checking, and password hashing.

2. Problem Definition

Implement a secure registration system that accepts user details, validates the submitted information, checks for duplicate email addresses and usernames, hashes passwords using PHP's password_hash() function, and stores the registration data in a MySQL database.

3. Objectives

- To connect PHP with a MySQL database using MySQLi.
- To validate user inputs on the frontend and backend.
- To check duplicate email addresses and usernames.
- To store passwords securely using password hashing.
- To insert valid user details into the database.
- To display appropriate success and error messages.

4. Software and Hardware Requirements

Software Requirements:

- XAMPP Server
- PHP
- MySQL
- phpMyAdmin
- Visual Studio Code
- Google Chrome
- HTML, CSS, and JavaScript

Hardware Requirements:

- Computer or laptop
- Minimum 4 GB RAM

5. Database Description

The practical uses a MySQL database to store registered user details.

The database table contains fields such as:

- id: Unique identification number.
- name: Name of the user.
- username: Unique username.
- email: Email address of the user.
- password: Hashed password.

The exact column names depend on the database table created for the practical.

6. Implementation Steps

Step 1: Start XAMPP

Start Apache and MySQL from the XAMPP Control Panel.

Step 2: Open phpMyAdmin

Open http://localhost/phpmyadmin in the browser and select the database created for the StudentWebsite project.

Step 3: Create or Verify the Database Table

Create or verify the registration table with the required fields for user information. Configure the username and email fields to prevent duplicate entries where appropriate.

Step 4: Create the Project Files

Create the required PHP, HTML, CSS, and JavaScript files inside the project directory.

Step 5: Establish the Database Connection

Use PHP MySQLi to connect the application to the MySQL database.

Step 6: Create the Registration Form

Design a registration form that accepts the user's name, username, email address, and password.

Step 7: Perform Frontend Validation

Use HTML validation attributes and JavaScript to check required fields, email format, and password input before submission.

Step 8: Perform Backend Validation

Validate the submitted data again using PHP before performing database operations. Frontend validation alone is not sufficient because it can be bypassed.

Step 9: Check Duplicate Email and Username

Query the database using prepared statements to check whether the submitted email address or username already exists. Display an appropriate message if a duplicate is found.

Step 10: Hash the Password

Use PHP's password_hash() function to generate a secure password hash before storing the password in the database. Never store the original plaintext password.

Step 11: Insert User Details

Use a MySQLi prepared statement to insert the validated user details and hashed password into the database.

Step 12: Display the Result

Display a success message when registration is completed or an appropriate error message when validation or database insertion fails.

Step 13: Test the Registration System

Test the application using valid details, invalid email addresses, empty fields, duplicate usernames, duplicate email addresses, and different password inputs.

7. Program Files

The practical may contain the following files, depending on the implementation:

1. db.php – Database connection.
2. register.php – Registration form and processing logic.
3. style.css – Styling for the registration interface.
4. script.js – Frontend validation, if implemented.

8. Output Screenshots

User Registration

This email is already registered.

This username is already taken.

Name

DDDDDrrrrrr

Username

25dce097

Email

drashti@gmail.com

Mobile

9876543210

Course

BCA

Password

Confirm Password

Register

User Registration

This email is already registered.

This username is already taken.

Passwords do not match.

Name

Drashtipatel

Username

25dce097

Email

drashti@gmail.com

Mobile

9876543210

Course

BCA

Password

Confirm Password

Register

User Registration

This email is already registered.

Name

Drashtipatel

Username

25dce0097

Email

drashti@gmail.com

Mobile

9876543219

Course

BCA

Password

Confirm Password

Register

User Registration

Name must be 3-100 letters and spaces only.

Name

R1

Username

test_a1

Email

testa1@example.com

Mobile

9876543211

Course

BCA

Password

••••••••

Confirm Password

••••••••

Register

User Registration

Name must be 3-100 letters and spaces only.
Username must be 4-20 characters: letters, numbers, underscore.

Name

R1

Username

ab

Email

testa1@example.com

Mobile

9876543211

Course

BCA

Password

••••••••

Confirm Password

••••••••

Register

User Registration

Enter a valid email address.

Name

Drashti Patel

Username

test_a3

Email

drashti.com

Mobile

9876543213

Course

BCA

Password

••••••••

Confirm Password

••••••••

Register

User Registration

Mobile number must be exactly 10 digits.

Name

Drashti Patel

Username

test_a4

Email

testa4@example.com

Mobile

98765

Course

BCA

Password

Confirm Password

Register

User Registration

Course is required.

Name

Drashti Patel

Username

test_a5

Email

testa5@example.com

Mobile

9876543215

Course

-- Select --

Password

•••••••••

Confirm Password

•••••••••

Register

User Registration

Password must be 8+ characters with uppercase, lowercase and a number.

Name

Drashti Patel

Username

test_a6

Email

testa6@example.com

Mobile

9876543216

Course

BCA

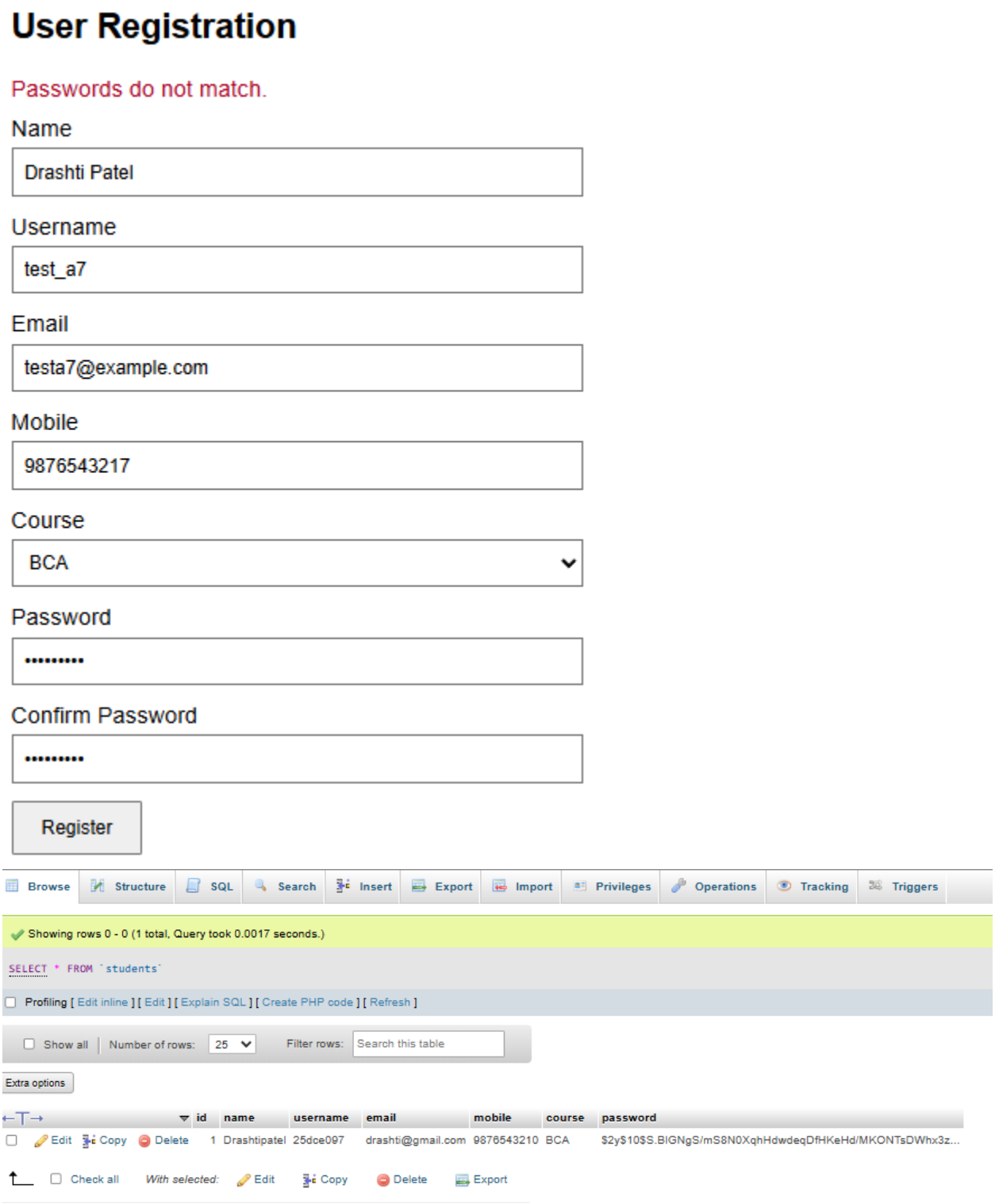
Password

••••••

Confirm Password

••••••

Register



Screenshot 1: Registration Form

[Insert screenshot here]

9. Testing and Results

The registration system is tested using different input conditions.

- Valid user details are accepted.
- Empty or invalid inputs are rejected.
- Duplicate email addresses and usernames are detected.
- Passwords are hashed before being stored.
- Valid user details are inserted into the MySQL database.
- Appropriate success and error messages are displayed.

The system helps protect registration data through server-side validation, prepared statements, duplicate checking, and password hashing.

10. Key Questions and Answers

Q1. Is password_hash() used before storing passwords?

Yes. The function generates a secure password hash that is stored instead of the original password.

Q2. Is duplicate email and username checking implemented?

Yes. The database is checked before inserting a new user record.

12. Learning Outcome

Students learn to implement secure user registration using PHP and MySQLi, validate input on both the frontend and backend, detect duplicate email addresses and usernames, hash passwords securely, and store user details in a database.

13. Conclusion

The secure user registration system was developed using PHP, MySQLi, and frontend technologies. It validates user input, checks for duplicate email addresses and usernames, hashes passwords using password_hash(), and stores valid user details in a MySQL database. The practical demonstrates essential techniques for developing safer registration systems.

Practical 10: Secure Login/Logout with Sessions, Role-Based Access, Timeout, and Remember-Me Token

1. Problem Definition

Develop secure login and logout functionality using PHP sessions. Verify the password using password_verify(), regenerate the session ID after login, restrict dashboard access, implement role-based redirection, session timeout, and an optional secure remember-me token.

2. Aim

To implement login and logout with secure session management, so that only authenticated users can reach protected pages and each role (admin or student) lands on the correct dashboard.

Database Design

Table: users

Column	Purpose
id	Primary key
name	User's full name
email	Unique login ID
password_hash	Hash created by password_hash()
role	admin or student
last_login	Time of the last successful login

Table: remember_tokens

Column	Purpose
id	Primary key
user_id	Linked user
selector	Random public identifier stored in the cookie
token_hash	SHA-256 hash of the random token
expires_at	Token expiry date

Seeded test users

Email	Password	Role
admin@example.com	Admin@123	admin
student@example.com	Student@123	student

ITUE:203-WDF **25DCE097**

Files Created

- db.php: database connection
- seed.php: inserts the seeded users with hashed passwords
- auth.php: checks login, timeout, remember-me cookie and role on every protected page
- login.php: login form and authentication
- admin_dashboard.php: admin only
- student_dashboard.php: student only
- logout.php: secure logout

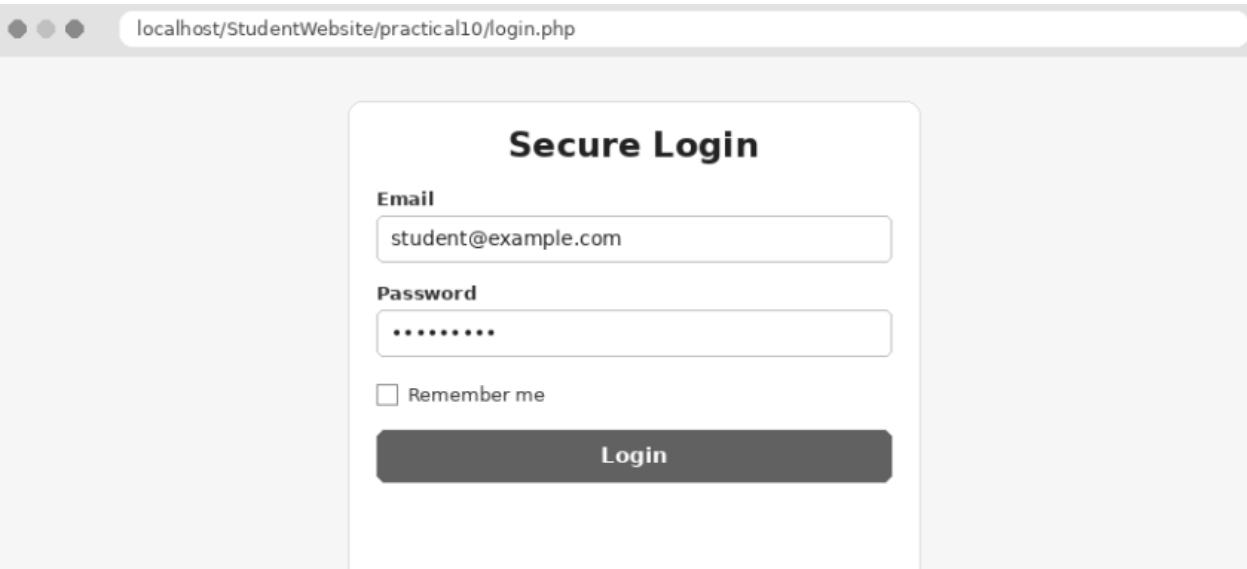
Test Cases

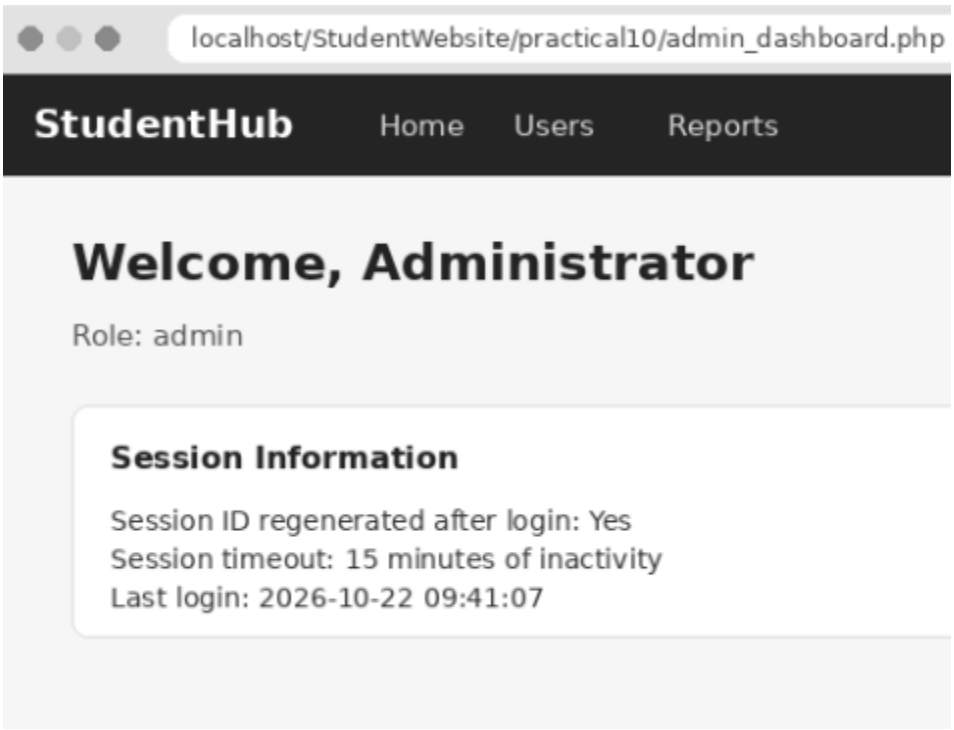
No.	Test	Expected Result	Result
1	Login as admin with correct password	Redirect to admin dashboard	Pass
2	Login as student with correct password	Redirect to student dashboard	Pass
3	Login with wrong password	Error message shown	Pass
4	Open dashboard URL without login	Redirect to login page	Pass
5	Student opens admin dashboard	Blocked, redirected to login	Pass
6	Stay idle past timeout, then refresh	Session expired message	Pass
7	Logout, then press the Back button	Dashboard not accessible	Pass
8	Tick Remember me, close and reopen browser	Auto-login works	Pass
9	Compare PHPSESSID before and after login	Session ID changes	Pass

Q. Why use password_hash() and password_verify()?
password_hash() creates a salted, slow hash, and password_verify() checks a password against it safely. Even if the database leaks, the real passwords are not exposed.

Q. Why regenerate the session ID after login?
To prevent session fixation, where an attacker plants a known session ID before the victim logs in.

Q. How is session timeout implemented?
The time of the last activity is saved in the session. On every request it is compared with the current time, and if the gap is over 15 minutes the session is destroyed.


localhost/StudentWebsite/practical10/login.php
Secure Login
Email
student@example.com
Password
Remember me
Login


localhost/StudentWebsite/practical10/admin_dashboard.php
StudentHub
Home
Users
Reports
Welcome, Administrator
Role: admin
Session Information
Session ID regenerated after login: Yes
Session timeout: 15 minutes of inactivity
Last login: 2026-10-22 09:41:07



Conclusion

Secure login and logout were implemented using PHP sessions, password_verify(), session ID regeneration, role-based redirection, idle timeout, a last-login timestamp, and a hashed remember-me token. All test cases passed.

Practical 11 :

Problem Definition : Create a student management module to add, view, edit, delete, search and

filter student records. All database operations must use MYSQLi

statements and clear success/failure messages.

Student Management System

Add, view, edit, delete, search, and filter student records.

Add New Student

Student name | Email address | Course | Mobile number | Add Student

Search and Filter Students

Search name, email, or cou | All Courses | Search / Filter | Reset

Student Records (20 found)

ID	Name	Email	Course	Mobile	Actions
20	Sara Joshi	sara@gmail.com	Information Technology	9876500020	Edit Delete
19	Om Desai	om@gmail.com	Civil Engineering	9876500019	Edit Delete

Student Records (20 found)

ID	Name	Email	Course	Mobile	Actions
20	Sara Joshi	sara@gmail.com	Information Technology	9876500020	Edit Delete
19	Om Desai	om@gmail.com	Civil Engineering	9876500019	Edit Delete
18	Jiya Mehta	jiya@gmail.com	Computer Engineering	9876500018	Edit Delete
17	Manav Shah	manav@gmail.com	Information Technology	9876500017	Edit Delete
16	Isha Patel	isha@gmail.com	Mechanical Engineering	9876500016	Edit Delete

Page 1 of 4 Next

Student Management System

Add, view, edit, delete, search, and filter student records.

Add New Student

Student name | Email address | Course | Mobile number | Add Student

Search and Filter Students

jiya | Computer Engineering | Search / Filter | Reset

Student Records (1 found)

ID	Name	Email	Course	Mobile	Actions
18	Jiya Mehta	jiya@gmail.com	Computer Engineering	9876500018	Edit Delete

Page 1 of 1

Activa
Go to S

Student Management System

Add, view, edit, delete, search, and filter student records.

Add New Student

drashti | 123qe@gmail.com | Civil Engineering | 1234567899 | Add Student

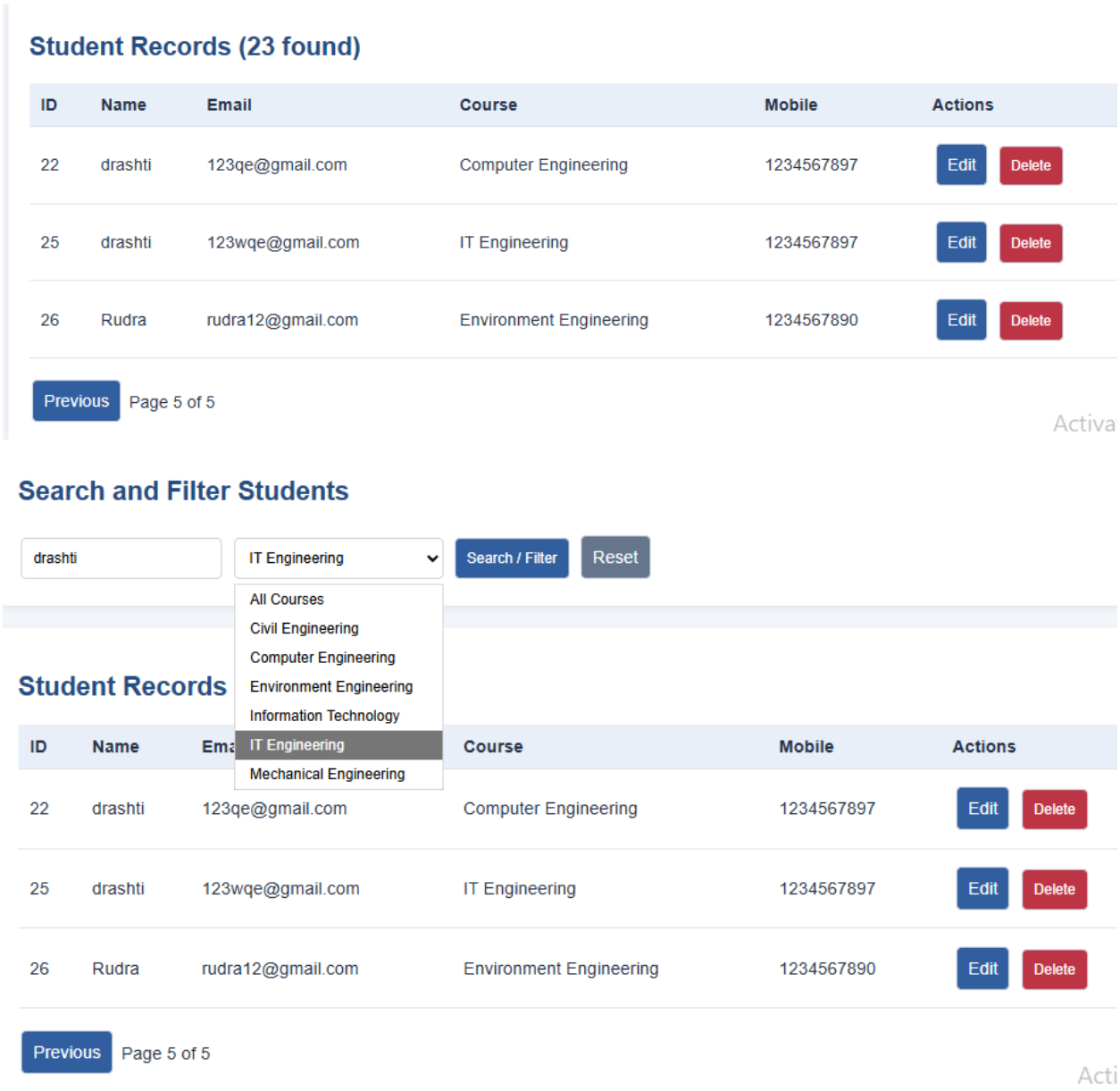
Search and Filter Students

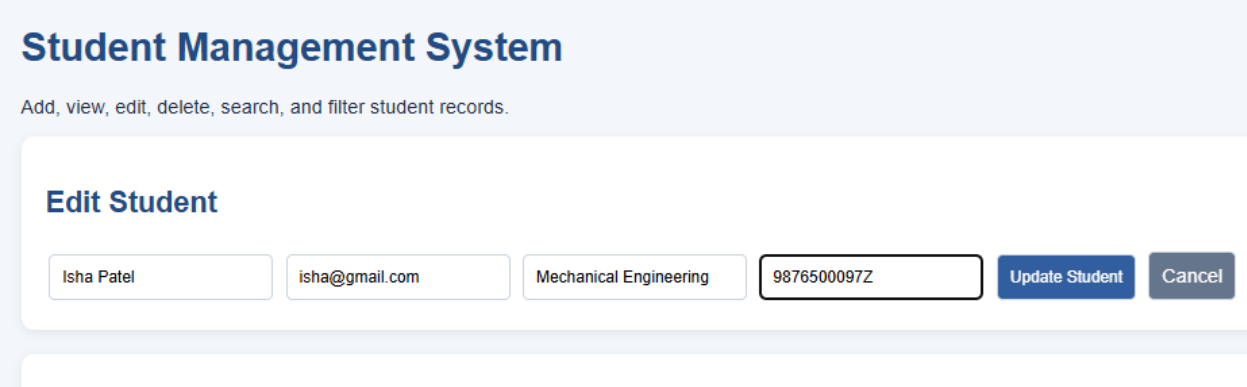
Search name, email, or cou | All Courses | Search / Filter | Reset

Student Records (21 found)

ID	Name	Email	Course	Mobile	Actions
22	drashti	123qe@gmail.com	Computer Engineering	1234567897	Edit Delete

Fatal error: Uncaught mysqli_sql_exception: Duplicate entry '123qe@gmail.com' for key 'email' in C:\xampp\htdocs\StudentWebsite\practical11\index.php:33 Stack trace: #0 C:\xampp\htdocs\StudentWebsite\practical11\index.php(33): mysqli_stmt->execute() #1 {main} thrown in **C:\xampp\htdocs\StudentWebsite\practical11\index.php** on line **33**





Step 10: Export the SQL database dump

In phpMyAdmin:

1. Select the student_management database.
2. Click Export.
3. Choose Quick and format SQL.
4. Click Export or Go, depending on your phpMyAdmin version.
5. Save the downloaded .sql file.

This file contains the database structure and records needed to recreate your database.

Step 11: Prepare the final submission

Your submission folder should contain:

Practical11_StudentManagement

db.php

index.php

export.php (if implementing CSV)

student_management.sql

screenshots/

test_cases.docx or test_cases.pdf

Test cases to submit

You can include this table in your practical report.

Test case	Input/action	Expected outcome
TC01	Add valid student	Student inserted
TC02	View student list	Records displayed
TC03	Update student course	Course updated
TC04	Delete student and confirm	Student removed
TC05	Search by name/email	Matching records displayed
TC06	Filter by course	Selected course displayed
TC07	Submit duplicate email	Error message displayed
TC08	Search for nonexistent student	No records found
TC09	Navigate pages	Correct records shown per page
TC10	Export CSV	CSV file downloaded

Record the actual result for each test after running it. Don't mark tests as passed until you've verified them.

Purpose and conclusion for the practical file

Purpose:

To implement a Student Management System using PHP and MySQLi that performs CRUD operations (Create, Read, Update, Delete), provides search and filtering by name, email, and course, and displays clear success and failure messages. The system also uses prepared statements for database queries, confirms deletion before execution, and supports pagination for managing student records efficiently.

Conclusion:

The Student Management System was implemented using PHP, MySQL, and MySQLi. It supports adding, viewing, editing, and deleting student records, searching and filtering data, and displaying paginated results. Prepared statements, input validation, and delete confirmation help improve the safety and usability of the application. CSV export can be included as an additional feature.

Before submitting: Test the full application in XAMPP, verify that at least 20 records exist, and confirm that add, edit, delete, search, filter, and pagination work. The code is not proof of a working practical until those operations have actually run.

Step 4: Write the database connection code Open db.php and paste this code: Save the file using Ctrl + S. This connects PHP to your existing MySQL database using MySQLi. The default XAMPP configuration usually uses username root with an empty password. ??? What is last part

The last part means that XAMPP usually uses these database login details by default:

- Username: root
- Password: Leave it empty
- Database name: student_management

In your db.php file, this line:

$conn = new mysqli("localhost", "root", "", "student_management");

means:

Code	Meaning
localhost	MySQL is running on your own computer
root	MySQL username
""	Empty password
student_management	Name of your database

What you need to do: If you haven't changed your MySQL password in XAMPP, leave the code as it is and save the file using Ctrl + S.

If you've set a MySQL password, replace the empty "" with your actual password.

You don't need to type anything extra. The empty quotes are intentional, not a missing piece of code.

Practical 12: Event Management System Using PHP and MySQL

1. Title

Event Management System Using PHP and MySQL

2. Problem Definition

Develop an Event Management System using PHP and MySQL that allows users to add, view, edit, and delete event details. The system stores event information in a database and displays the available events on a webpage.

The system also supports event categories, dates, descriptions, poster uploads, and event status. Events that have existing registrations cannot be deleted, ensuring database consistency.

This practical is implemented as a standalone module and does not need to be integrated into the Student Hub Portal.

3. Objectives

- To understand database connectivity using PHP and MySQL.
- To perform CRUD operations on event records.
- To insert, display, update, and delete event details.
- To upload and manage event poster images.
- To validate user input and handle errors.
- To use prepared statements for safer database operations.
- To maintain database integrity when events have registrations.

4. Software and Technologies Used

- Frontend: HTML and CSS
- Backend: PHP
- Database: MySQL
- Database Management Tool: phpMyAdmin
- Local Server: XAMPP
- Web Browser: Google Chrome

5. Database Design

Database name: studenthub

Table name: events

Field Name	Description
id	Unique ID of each event
event_name	Name of the event
event_date	Date of the event
description	Details about the event
category	Category of the event
poster	Event poster image filename
status	Event status: open or closed

The id field uniquely identifies each event. The status field indicates whether an event is open or closed.

6. Implementation Description

The Event Management System is implemented using separate PHP files for different operations.

6.1 Database Connection

The db.php file establishes a connection between PHP and the MySQL database. It allows the application to perform database operations.

6.2 Display Events

The index.php file retrieves event records from the database and displays them in a list or table. It also provides options to add, edit, and delete events.

6.3 Add Event

The add_event.php file allows users to enter event details, including the event name, date, description, category, status, and optional poster image. The submitted details are validated and stored in the database.

6.4 Edit Event

The edit_event.php file allows users to modify existing event details. The selected event information is retrieved using its ID, and the updated information is saved in the database.

6.5 Delete Event

The delete_event.php file handles event deletion. Before deleting an event, the system checks whether registrations are associated with it. If registrations exist, deletion is prevented to maintain database integrity.

6.6 Input Validation and Security

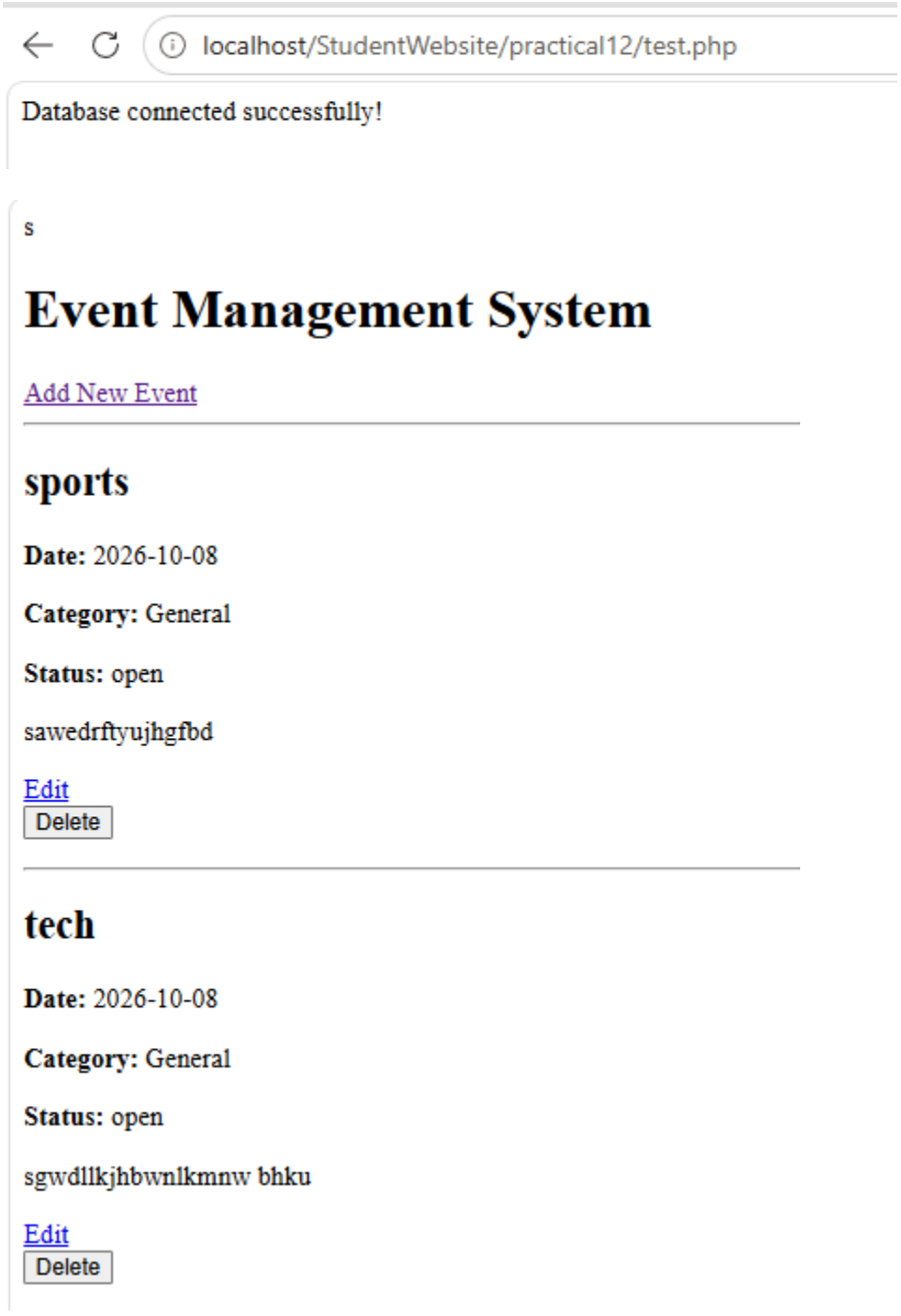
The application uses input validation to check the submitted information. Poster uploads are restricted by file type and size. Prepared SQL statements are used for database operations to reduce the risk of SQL injection.

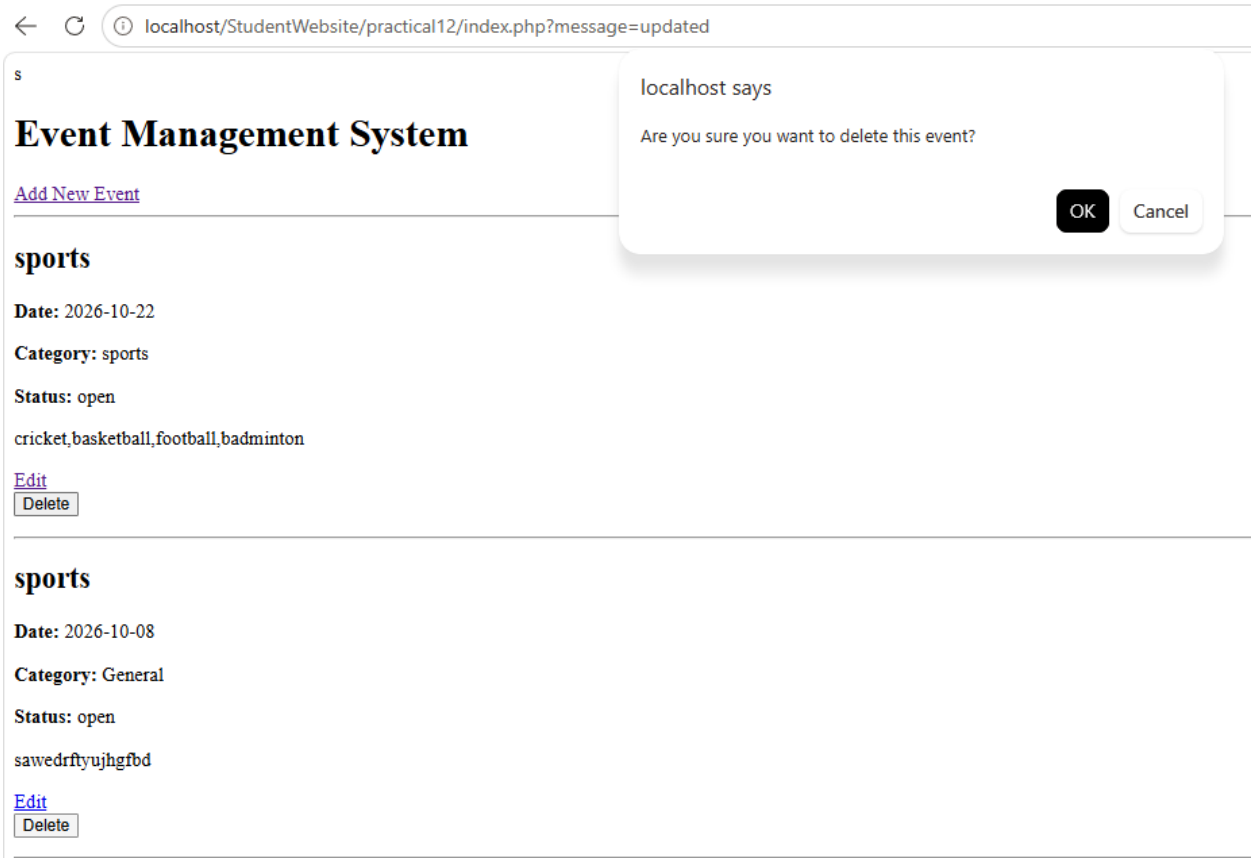
7. Execution Procedure

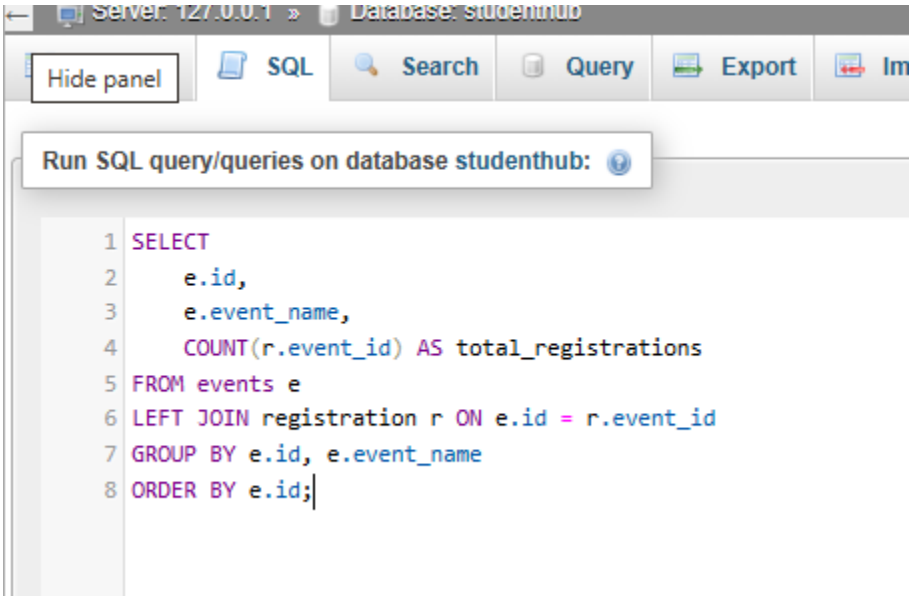
1. Start Apache and MySQL using the XAMPP Control Panel.
2. Open phpMyAdmin and verify that the studenthub database and events table exist.
3. Save the practical files in C:\xampp\htdocs\StudentWebsite\practical12.
4. Open the browser and visit http://localhost/StudentWebsite/practical12/index.php.
5. Verify that the event list is displayed.
6. Add a new event and check whether it appears in the event list.

7. Edit an existing event and verify that the changes are saved.
8. Test poster upload functionality, if required.
9. Attempt to delete an event and verify the system's response.
10. Confirm that events with existing registrations cannot be deleted.

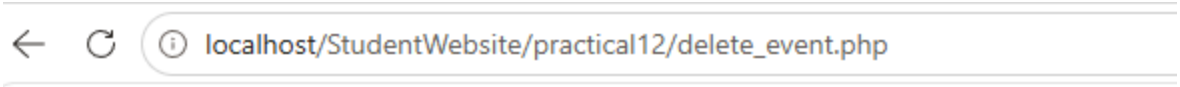
8. Output Screenshots




localhost/StudentWebsite/practical12/index.php?message=updated
s
Event Management System
Add New Event
localhost says
Are you sure you want to delete this event?
OK
Cancel
sports
Date: 2026-10-22
Category: sports
Status: open
cricket,basketball,football,badminton
Edit
Delete
sports
Date: 2026-10-08
Category: General
Status: open
sawedrftyujhgfbd
Edit
Delete


Hide panel
SQL
Search
Query
Export
Run SQL query/queries on database studenthub:
SELECT
e.id,
e.event_name,
COUNT(r.event_id) AS total_registrations
FROM events e
LEFT JOIN registration r ON e.id = r.event_id
GROUP BY e.id, e.event_name
ORDER BY e.id;


Browse
Structure
SQL
Show query box
Current selection does not contain a unique column.
Showing rows 0 - 2 (3 total, Query took 0.0035 secon
SELECT e.id, e.event_name, COUNT(r.event_id)
e.id;
Profiling [Edit inline] [Edit] [Explain SQL] [Create
Show all
Number of rows:
25
Extra options
id 1
event_name
total_registrations
1 tech 1
211 sports 1
212 sports 0



sCannot delete this event because registrations exist. Keep the event or handle its registrations first.

s

Event Management System

Add New Event

---

sports

Date: 2026-10-30

Category: General

Status: open

Badminton,kho-kho,football

Edit
Delete

---

sports

Date: 2026-10-08

Category: General

Status: open

sawedrftyujhgfbd

Edit
Delete

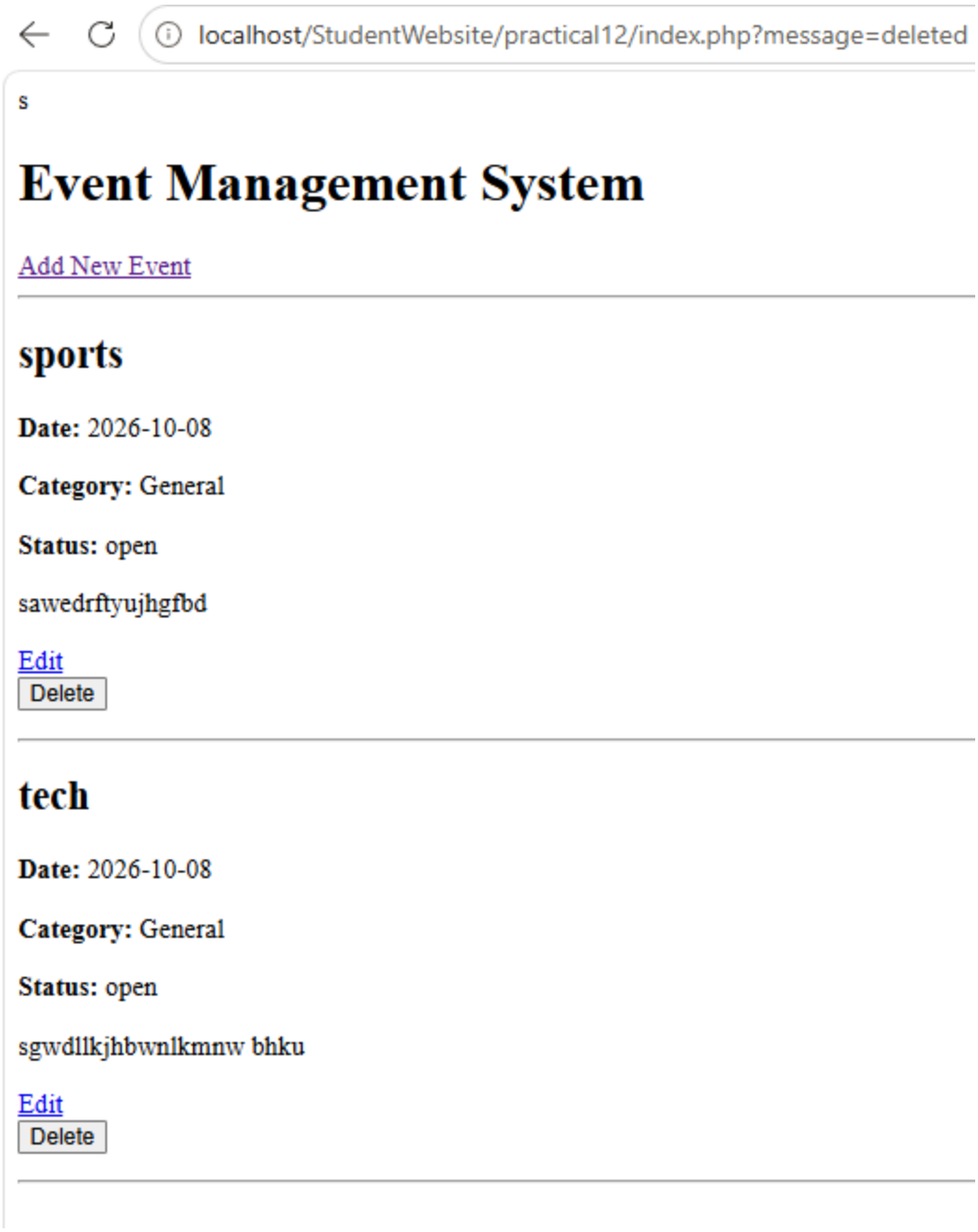
---

tech

Date: 2026-10-08

Category: General

Status: open


localhost/StudentWebsite/practical12/index.php?message=deleted
s
Event Management System
Add New Event
sports
Date: 2026-10-08
Category: General
Status: open
sawedrftyujhgfbd
Edit
Delete
tech
Date: 2026-10-08
Category: General
Status: open
sgwdllkjhbwnlkmnw bhku
Edit
Delete



9. Result

The Event Management System was developed successfully using PHP and MySQL. The application supports creating, viewing, updating, and deleting event records. It also provides poster upload functionality, input validation, and protection against deleting events that have existing registrations.

The practical demonstrates the use of database connectivity, CRUD operations, prepared statements, and database integrity in a PHP-based web application.

10. Conclusion

This practical helped demonstrate how PHP interacts with a MySQL database to manage event information. It provided practical experience in performing CRUD operations, validating user input, handling image uploads, and maintaining

relationships between database tables. The completed module can operate independently without integration into the Student Hub Portal.

Practical 13: PHP JSON API and AJAX/Fetch-Based CRUD Without Page Reload

1. Aim

To develop a PHP JSON API connected to a MySQL database and perform Create, Read, Update, and Delete (CRUD) operations using JavaScript Fetch API without reloading the webpage.

2. Problem Definition

Create PHP JSON API endpoints for students and consume them using the Fetch API. Perform listing, adding, updating, and deleting operations asynchronously. Display appropriate loading, success, and error messages on the frontend.

3. Objectives

- To understand PHP JSON APIs.
- To connect PHP with a MySQL database.
- To implement asynchronous CRUD operations using Fetch API.
- To exchange data between the frontend and backend using JSON.
- To validate user input and handle errors.
- To inspect API requests and responses using the browser Network tab.

4. Software and Hardware Requirements

Software Requirements:

- XAMPP Server
- PHP
- MySQL
- Visual Studio Code
- Google Chrome
- HTML, CSS, and JavaScript

Hardware Requirements:

- Computer or laptop
- Minimum 4 GB RAM
- Internet browser

5. Database Description

The practical uses a MySQL database containing a students table.

The table contains the following fields:

- id: Unique student identification number.
- name: Name of the student.
- email: Email address of the student.
- mobile: Mobile number of the student.
- course: Course selected by the student.

6. Implementation Steps

Step 1: Start XAMPP

Start Apache and MySQL from the XAMPP Control Panel.

Step 2: Create the Project Folder

Create a folder named practical13 inside the StudentWebsite directory located in C:\xampp\htdocs.

Step 3: Create the Database Connection

Create db.php to establish a connection between PHP and the MySQL database.

Step 4: Create the PHP API

Create api.php to handle GET and POST requests, perform database operations, validate inputs, and return JSON responses.

Step 5: Test the API Endpoint

Open the API URL in the browser and verify that the endpoint returns a valid JSON response containing student records.

Step 6: Create the Frontend Interface

Create index.php containing a student registration form and a table to display student records.

Step 7: Apply CSS Styling

Create style.css to improve the appearance of the form, buttons, messages, and student table.

Step 8: Implement Fetch API

Create script.js to send asynchronous requests to the PHP API and process JSON responses.

Step 9: Perform CRUD Operations

Implement the following operations:

- **Create:** Add a new student to the database.
- **Read:** Retrieve and display student records.
- **Update:** Modify an existing student's details.
- **Delete:** Remove a student record from the database.

These operations are performed without a full-page reload.

Step 10: Test the Network Tab

Open Developer Tools in Google Chrome and select the Network tab. Use the Fetch/XHR filter to monitor requests sent to api.php.

Step 11: Inspect API Responses

Select the api.php request and inspect its Headers, Payload, and Response tabs to verify the request method, submitted data, and JSON response.

Step 12: Validate Input and Handle Errors

Test empty fields, invalid email addresses, incorrect mobile numbers, and unsuccessful requests. Verify that appropriate error messages are displayed.

7. Program Files

The practical contains the following files:

1. index.php – Frontend interface.
2. db.php – Database connection.
3. api.php – PHP JSON API endpoints.
4. script.js – Fetch API and CRUD operations.
5. style.css – Frontend styling.

8. Output Screenshots

localhost/StudentWebsite/practical13/api.php

Pretty-print

{"success":true,"message":"Students fetched successfully","data":
[{"id":"1","name":"abc","email":"123abc@gmail.com","mobile":"1234567890","course":"computer"},
{"id":"2","name":"wqwq","email":"abcd12@yahoo.com","mobile":"9087654321","course":"web"},
{"id":"3","name":"fssfeg","email":"123aa@gmail.com","mobile":"1234567890","course":"computer"},
{"id":"4","name":"srghfyj","email":"abc12@gmail.com","mobile":"2134567890","course":"computer"},
{"id":"5","name":"drashti","email":"123qe@gmail.com","mobile":"1239567890","course":"Computer Science"},
{"id":"6","name":"drashti","email":"123qe@gmail.com","mobile":"1239567890","course":"Computer
Engineering"}]}

Student Management

Name

drashti

Email

123de@gmail.com

Mobile

9876543210

Course

Mechanical Engineering

Add Student

Cancel Edit

Student Records

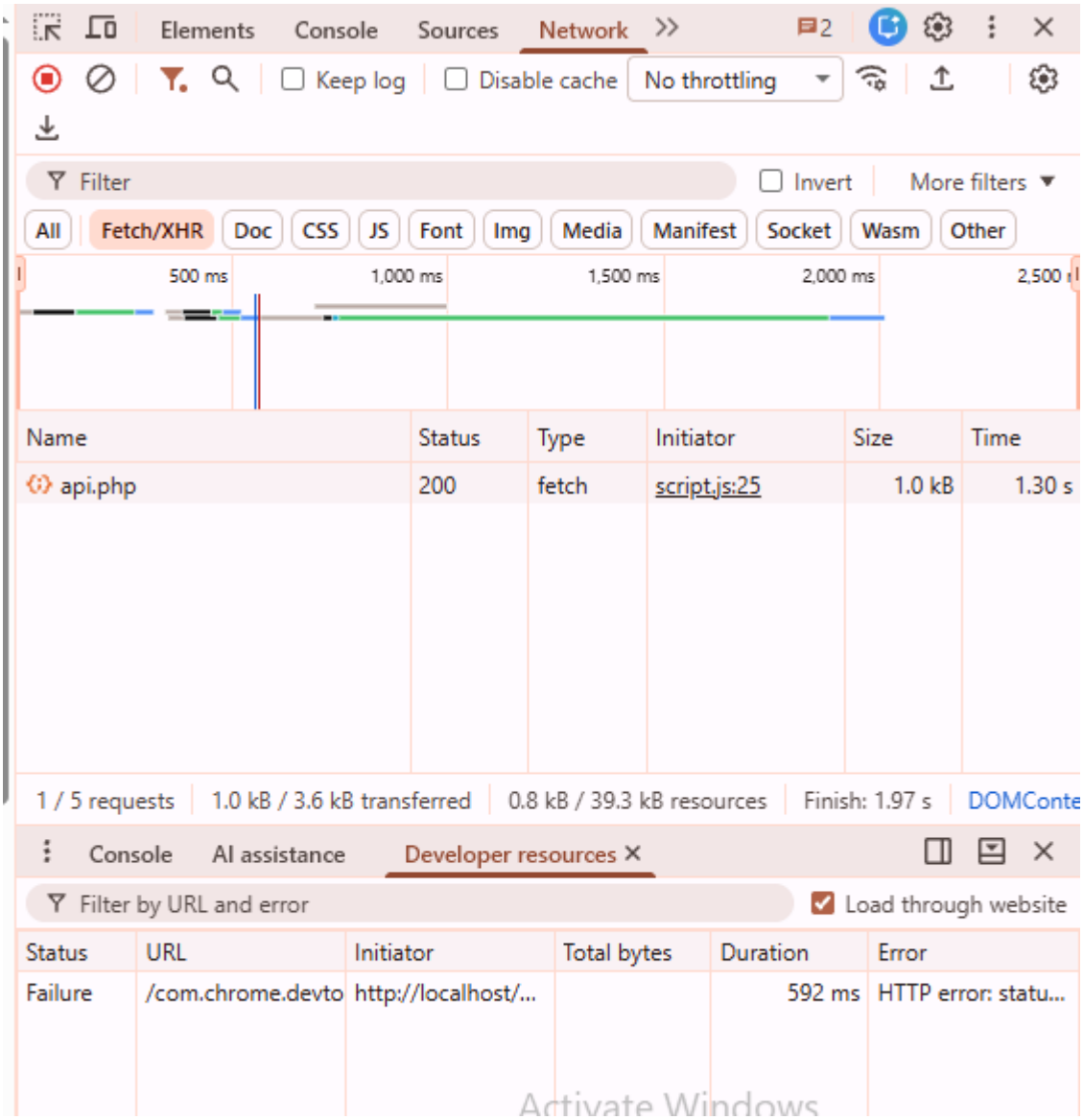
ID	Name	Email	Mobile	Course	Actions
1	abc	123abc@gmail.com	1234567890	computer	Edit Delete
2	wqwq	abcd12@yahoo.com	9087654321	web	Edit Delete
3	fssfeg	123aa@gmail.com	1234567890	computer	Edit Delete
4	srghfyj	abc12@gmail.com	2134567890	computer	Edit Delete

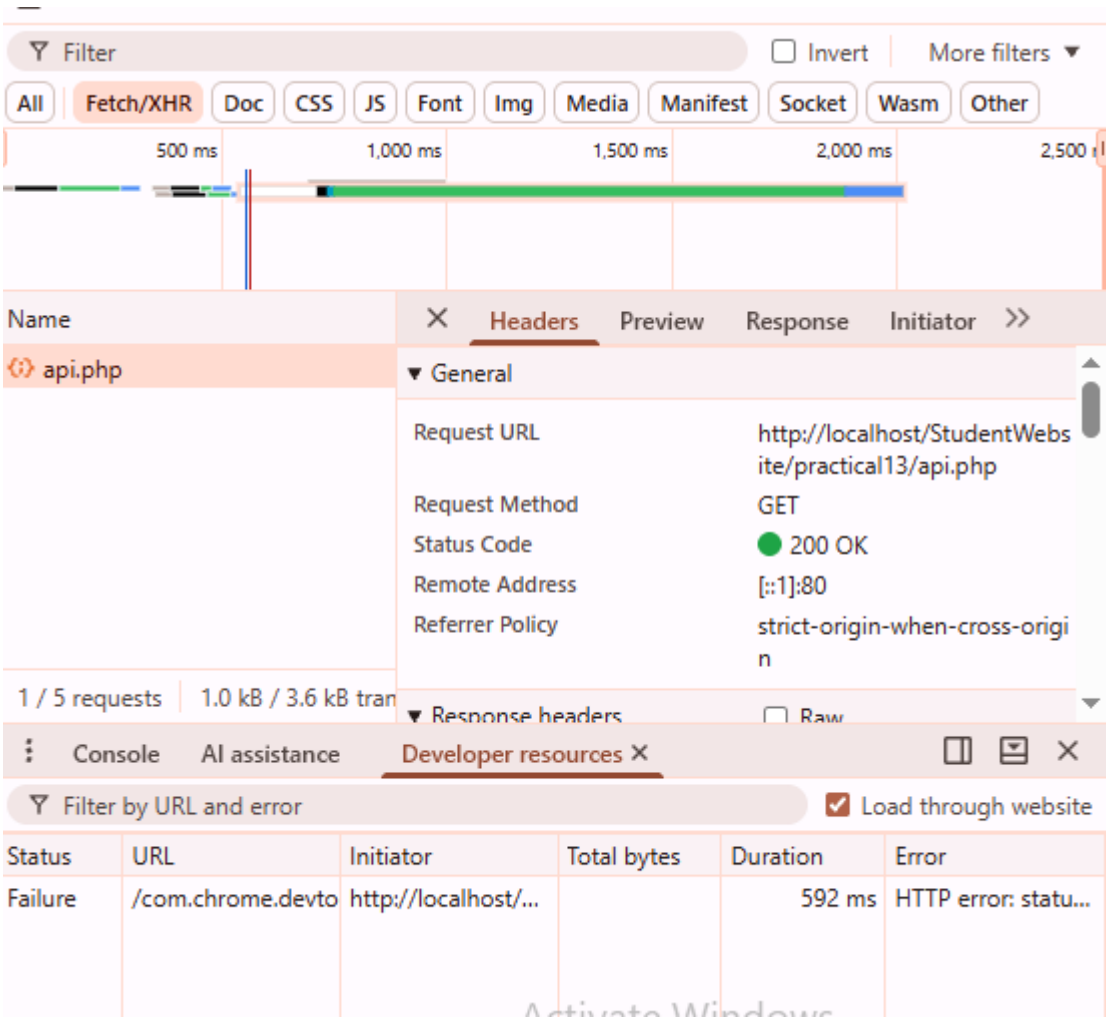
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Student Records

ID	Name	Email	Mobile	Course	Actions
1	abc	123abc@gmail.com	1234567890	computer	Edit Delete
2	wqwq	abcd12@yahoo.com	9087654321	web	Edit Delete
3	fssfeg	123aa@gmail.com	1234567890	computer	Edit Delete
4	srghfyj	abc12@gmail.com	2134567890	computer	Edit Delete
5	drashti	123qe@gmail.com	1239567890	Computer Science	Edit Delete
6	drashti	123qe@gmail.com	1239567890	Computer Engineering	Edit Delete
7	drashti	123de@gmail.com	9876543210	Mechanical Engineering	Edit Delete

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9. Testing and Results

The application is tested using different inputs and CRUD operations.

- Student records are retrieved from the MySQL database.
- New student records are added successfully.

- Existing student details are updated.
- Student records are deleted.
- API requests and JSON responses are inspected through the browser Network tab.
- Input validation and error messages are tested.

The application updates the displayed records without reloading the entire webpage.

10. Key Questions and Answers

Q1. Do PHP endpoints return valid JSON responses?

Yes. The PHP API uses json_encode() to return data in JSON format.

Q2. Are GET and POST requests handled properly?

Yes. GET retrieves student records, while POST handles adding, updating, and deleting records.

Q3. Are loading, success, and error states shown on the frontend?

Yes. JavaScript displays loading indicators and appropriate success or error messages.

Q4. Is input validation performed before database operations?

Yes. Required fields, email addresses, mobile numbers, and student IDs are validated before database modifications.

11. Applications

- Student management systems.
- Online registration portals.
- Interactive dashboards.
- Event management systems.
- API-driven web applications.
- Applications requiring asynchronous database operations.

12. Learning Outcome

Students learn to integrate JavaScript Fetch API with PHP and MySQL, exchange data using JSON, implement asynchronous CRUD operations, validate user inputs, and inspect API requests using browser Developer Tools.

13. Conclusion

The PHP JSON API and Fetch API-based CRUD application was implemented using PHP, MySQL, HTML, CSS, and JavaScript. Student records can be added, viewed, updated, and deleted asynchronously without a full-page reload. The practical demonstrates frontend-backend integration, JSON data exchange, input validation, and API testing through the browser Network tab.

Practical 14: Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

1. Problem Definition

Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

2. Aim

To build a role-based admin dashboard with basic monitoring and traceability.

Database Design

Table	Main Columns
users	id, name, email, password_hash, role, status (active/inactive)
students	id, user_id, enrollment_no, department
events	id, title, event_date
registrations	id, student_id, event_id, registered_at
audit_logs	id, user_id, action, details, ip_address, created_at

Files Created

- guard.php: allows only logged-in admins and provides the audit logging function
- menu.php: dynamic menu based on role
- dashboard.php: statistics and latest registrations
- users.php: user list with active/inactive toggle
- toggle_status.php: changes user status and writes an audit log entry
- audit_logs.php: shows the audit log
- analytics.php: registrations by month chart

7. Key SQL Queries

Purpose	Query idea
Total users	COUNT of all rows in users
Student count	COUNT of users where role is student
Event count	COUNT of rows in events
Latest registrations	JOIN of registrations, students, users, events, ORDER BY registered_at DESC, LIMIT 5
Monthly analytics	COUNT of registrations, GROUP BY month

8. Access Control Test Cases

No.	Test	Expected Result	Result
1	Admin opens dashboard	Dashboard shown with live data	Pass
2	Student opens dashboard URL	403 Forbidden	Pass
3	Visitor not logged in opens dashboard	Redirect to login	Pass
4	Admin menu	All admin options shown	Pass
5	Student menu	Only student options shown	Pass

9. Audit Log Test Cases

No.	Admin Action	Expected audit_logs Entry	Result
1	Open dashboard	VIEW_DASHBOARD with user id, IP and time	Pass
2	Deactivate user 3	TOGGLE_STATUS: user 3 set to inactive	Pass
3	Activate user 3	TOGGLE_STATUS: user 3 set to active	Pass
4	Admin deactivates own account	Blocked, no change	Pass

Q. How are dashboard values loaded dynamically?
Each card runs a SQL COUNT query when the page loads, so the numbers always match the current database.

Q. How is access restricted to admins?
guard.php checks the session for a logged-in user whose role is admin on every admin page. Everyone else is redirected or shown a 403 message.

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Student	Email	Event	Date
Riya Patel	riya@example.com	Tech Fest 2026	2026-10-21
Amit Shah	amit@example.com	Hackathon	2026-10-20
Neha Joshi	neha@example.com	Workshop PHP	2026-10-19
Karan Mehta	karan@example.com	Tech Fest 2026	2026-10-18

Role: admin

ID	Admin	Action	Details	Time
5	admin	TOGGLE_STATUS	user #3 set to inactive	2026-10-22 10:14
4	admin	LOGIN	admin logged in	2026-10-22 10:10
3	admin	VIEW_DASHBOARD	dashboard opened	2026-10-22 10:10
2	admin	LOGOUT	admin logged out	2026-10-21 17:02
1	admin	TOGGLE_STATUS	user #3 set to active	2026-10-21 16:55

12. Conclusion

An admin-only dashboard was built with live statistics, role-based menus, a user status toggle, an audit trail for admin actions, and a monthly registrations chart. Access control and audit logging were verified with test cases.